

The LQD Model: Coordinates for the Space of Smiles

A density-first lecture on arbitrage-free volatility fitting

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Abstract

A volatility smile is a probability distribution seen through the Black formula. Most fitters nevertheless parametrize the smile directly and then police, with penalties, the awkward nonlinear conditions under which their curve still corresponds to a genuine distribution. This lecture develops the Vol-Fitter's *LQD* (log-quantile-density) model the other way around, as the answer to a coordinate problem: among all the charts one can draw on the space of arbitrage-free slices — price, implied volatility, distribution function, density, quantile — find the one in which that space becomes *all of coordinate space*, so that the optimizer can roam freely and every point it visits already prices from a genuine law. The answer is to model the logarithm of the quantile density of the log-forward return as two universal boundary terms plus a Legendre expansion, $\log q(u) = -\log u - \log(1-u) + g(u)$. Positivity of the density is then structural, a single scalar shift enforces the martingale condition, and the two endpoint values of g identify both the last finite moments of the distribution and its Lee wing slopes; the only genuine constraint that survives is one inequality, $A_R < 1$, the price of a finite forward. From this construction we derive pricing as one cumulative integral, exact closed-form ATM level/skew/curvature handles with an orthogonal chart around them, the calibration objective, an analytic residual Jacobian obtained from a clean cancellation identity, and soft convex-order control across expiries — read off the same upper-share ledger that prices the calls. Every figure is generated by the production calibrator on deterministic synthetic benchmarks: a ten-parameter fit to an SPX-like strip reaches a maximum error of 1.341 volatility basis points, and a seventeen-parameter fit recovers an asymmetric bimodal event density to 2.444 vol basis points. Market-fit evidence lives in the backtest reports; what this lecture establishes is the machine itself.

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1 The shape of the problem

Here is a failure mode every production smile fitter meets. You parametrize implied volatility with some pleasant curve, fit seven liquid strikes, and the residuals look excellent. Then a risk system differentiates your curve twice — because the second strike derivative of a call *is* the risk-neutral density, as we recall below — and finds it negative between two quotes. Your smooth, well-fitting smile is quoting a butterfly you would pay to be lifted on: free money for the counterparty, a phantom probability of -2% for the risk report. Nothing in the fit was wrong numerically. The parametrization simply did not know that a smile is not free to be any curve.

The standard remedy is to keep the pleasant curve and add penalties: evaluate the density (or its proxy) on a grid, punish negativity, and hope the optimizer lands inside the feasible set. This works, mostly, but it has a characteristic cost profile. The constraint is *global* and *nonlinear* in the parameters, so it must be re-checked at every strike and every iterate; the optimizer spends its time crawling along the boundary of the feasible set, exactly where the penalty gradients fight the data gradients; and when the market is sparse or stressed — precisely when you most need the fitter — the boundary is where the solution wants to live.

The LQD model spends none of that effort, because it asks a different question first.

The set of arbitrage-free slices is a well-defined mathematical object. In which coordinates is that set trivial — all of coordinate space, no boundary, nothing to police?

This lecture is the answer worked out in full, and the sense in which it is a lecture is worth one sentence: we will build the model from first principles, prove the small number of facts that carry it, and keep the numerical plumbing in clearly marked asides, so that a reader who knows the Black formula and one year of probability can reconstruct everything. The destination is fixed by production, and it is protected by the following invariants.

Invariant

1. Every admissible parameter vector prices from a genuine non-negative density: butterfly-freedom is structural, never a penalty (proposition 1).
2. The only hard admissibility condition is the single inequality $A_R < 1$ — a finite forward — guarded by a smooth barrier before the wall (sections 4 and 9).
3. The martingale normalization is one scalar shift; it changes neither the density's shape, nor its tails, nor its positivity (section 7).
4. The ATM handles $(\sigma_0, s_0, \kappa_0)$ are exact closed-form functionals of the slice — the same three coordinates the graph extrapolator of Note 14 propagates (section 8).

A running example. Lectures profit from a recurring character. Ours is a deterministic, SPX-like strip of 24 quotes at the expiry $\tau = 0.75$, sampled from a skewed SSVI-shaped curve (appendix C) with a reproducible sub-two-basis-point microstructure ripple, and fitted *once* by the production calibrator with a nine-mode LQD slice. Almost every figure in this note draws that one fitted slice from a different angle; its full case file, with numbers, is section 12.1. A second character — a bimodal event distribution — appears when we need to stress the basis (section 12.2).

What this note is not. The examples are *synthetic approximation benchmarks*: they measure how well the parametrization can reproduce a known clean curve, not how well it fits noisy market quotes. Market-fit evidence — RMS against bid–ask NBBO quotes, in and out of sample, against an SVI-JW baseline — lives in the backtest harness (`backend/backtest/`) and its reports.

2 What a smile really is

2.1 Normalized units, and the three clocks

Fix one expiry. All the securities we price are functions of the terminal spot S , and everything deterministic — discounting, forwards, carry — can be divided out before modelling begins. We therefore work with the *log-forward return*

$$X = \log \frac{S}{F}, \quad Y = e^X = \frac{S}{F}, \quad (1)$$

with F the forward to the expiry, under the forward measure of that expiry. Strikes are quoted in log-moneyness $k = \log(K/F)$, and the *normalized call* is

$$C(k) = \mathbb{E}[(e^X - e^k)^+], \quad \mathbb{E}[e^X] = 1, \quad (2)$$

where the second equality — the martingale, or forward, condition — says that the forward really is the mean of the terminal spot. A dollar quote $C^\$$ converts to these units by $C = C^\$ / (DF)$ with D the discount factor; nothing else about rates or carry will appear again. Deterministic carry is a standing assumption of the whole note; it is what legitimizes working slice by slice under each expiry’s forward measure, and it is revisited when slices are coupled in section 11.

One symbol of time is enough, but it must be the right one. Production distinguishes the *expiry date* from its *calendar year fraction* and from the *active variance time* — calendar time dilated by scheduled events (Note 11). Everything in this note that multiplies a variance, scales a vega, or annualizes a handle uses the active time, written

$$\tau = \text{active variance time to expiry} \quad (\text{the calibrator's } \texttt{prepared.tau}), \quad (3)$$

which equals the calendar fraction whenever the event clock is off. Total implied variance at log-moneyness k is $w(k) = \sigma^2(k) \tau$, with σ the Black implied volatility. A *volatility basis point* (vol bp) is 10^{-4} in volatility. When several expiries appear at once we index them $\tau_1 < \tau_2 < \dots$; a single unlabelled slice is always “the” expiry.

2.2 Breeden–Litzenberger, and the slice as a law

The reason a smile cannot be an arbitrary curve is one integration by parts away. Write the call as a function of the normalized strike $y = e^k$; when Y has a density f_Y ,

$$\frac{\partial C}{\partial y} = -\mathbb{P}(Y > y), \quad \frac{\partial^2 C}{\partial y^2} = f_Y(y) \geq 0. \quad (4)$$

This is the Breeden–Litzenberger identity [1]: the call curve determines the distribution, and conversely every property of the call curve that a static portfolio can monetize is a property of

the distribution. A *butterfly* — long one call at $y - h$, long one at $y + h$, short two at y — costs $C(y - h) - 2C(y) + C(y + h)$, and by (4) its price per h^2 converges to the density at y . A negative butterfly price is exactly a negative density.

Definition 1 (Arbitrage-free slice). A *slice* is a call curve $C(k)$ of the form (2) for some law of X with $\mathbb{E}[e^X] = 1$. Equivalently (for laws with a density): C is decreasing and convex in the strike $y = e^k$, with $C \rightarrow 1$ as $y \rightarrow 0$ and $C \rightarrow 0$ as $y \rightarrow \infty$, and slope ≥ -1 .

The space of slices, then, is really the space of mean-one probability laws on $(0, \infty)$. That is the object we must parametrize, and fig. 1 shows our running example wearing both descriptions at once: a convex call curve on the left, and on the right the density recovered by pricing actual finite-width butterflies on the fitted slice.

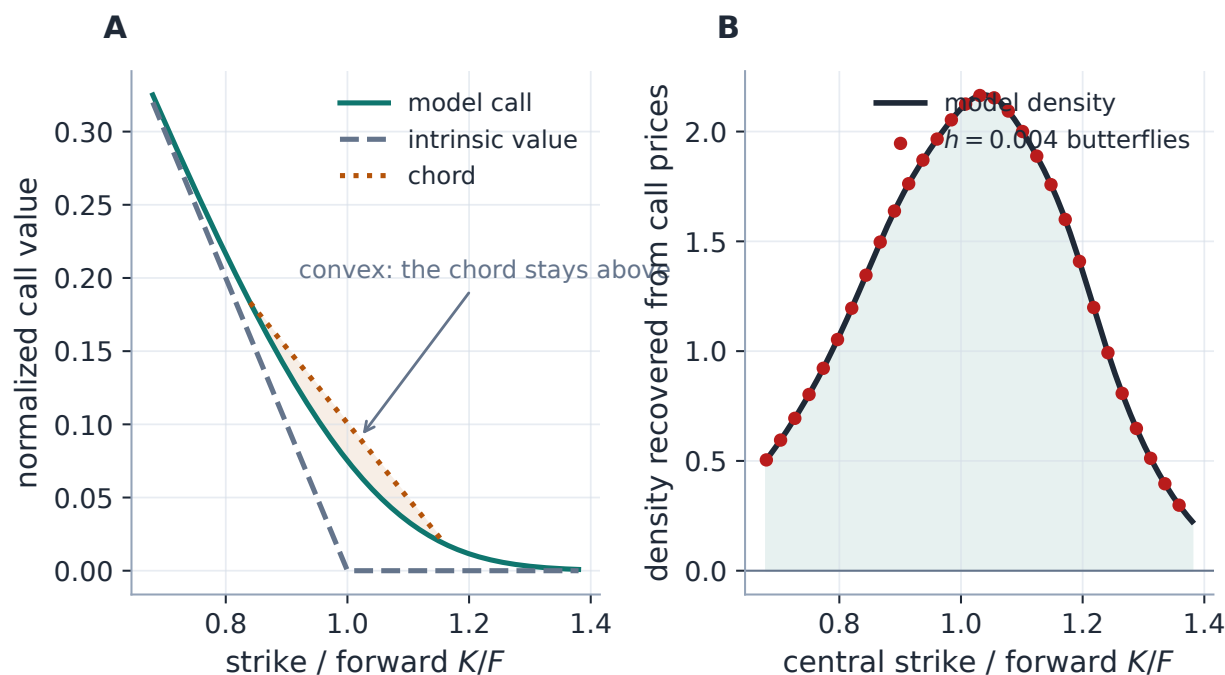


Figure 1: The running SPX-like fit as a tradable object. Panel A: the normalized call curve is decreasing and convex — every chord sits above the curve, so every butterfly has non-negative price. Panel B: discrete butterflies of half-width $h = 0.004$, priced off the fitted slice and divided by h^2 , recover the model density to a relative 0.0182 (the minimum butterfly value on the grid is $0.224428 > 0$). Butterfly-freedom is not something this fit achieved; it is something the parametrization cannot lose.

2.3 The notation ledger

Because this lecture insists on few symbols, here they all are. Each is defined again where it first works for a living; none is ever reused with a second meaning.

$X, Y = e^X$	log-forward return, gross return	$u, z = \log \frac{u}{1-u}$	rank (percentile), log-odds
$k, C(k), P(k)$	log-moneyness; call, put	$Q, q = Q'$	quantile of X ; quantile density
w, σ, τ	total variance $\sigma^2\tau$; vol; active time	g, θ	smooth part of $\log q$; coefficients
$B(k, w)$	normalized Black call	A_L, A_R	endpoint (tail) scales $e^{g(0)}, e^{g(1)}$
μ	martingale shift	G	upper share $\int e^Q$ above a rank
σ_0, s_0, κ_0	ATM vol, skew, curvature	β, ψ, p	Lee slope, Lee map, moment order
$\lambda, r; c, \gamma$	ridge; barrier centre/steepness	H, J, U, V, α, ξ	the ATM-orthogonal chart

Table 1: Every symbol in the note. Φ and φ are the standard normal CDF and PDF. One differentiation convention holds throughout: a *prime* is the derivative of a function of one variable (C', w', σ'); a *subscripted* ∂ is a partial derivative of a function of several ($\partial_k B, \partial G/\partial\theta$). Symbols local to one appendix (the SSVI oracle's ϑ, ρ, ϕ) stay in that appendix.

3 Six charts for one space

We are hunting for coordinates on the space of mean-one laws. A good way to appreciate the answer is to price the alternatives first — this is a shopping trip, and the currency is *constraints left over*.

The price chart. Parametrize $C(k)$ itself. The constraint set definition 1 is a convex cone with curved walls: monotonicity and convexity at *every* strike, plus the endpoint conditions. Convexity of a parametric curve is a global condition on second derivatives; enforcing it for a flexible family means a grid of inequality constraints. Nothing is free.

The volatility chart. Parametrize $\sigma(k)$ — what every trader quotes, and what SVI does. The density is now a *nonlinear* second-order differential expression in $w(k) = \sigma^2(k)\tau$ (Gatheral's g -function [3, 4]); non-negativity is an implicit, parameter-global boundary that no useful family satisfies identically. This chart is the most convenient to *quote* and the most expensive to *police* — the note's opening failure mode lives here.

The distribution chart. Parametrize the CDF F_X . One constraint survives: monotonicity, again at every point. Closer.

The density chart. Parametrize $f_X \geq 0$ directly. Positivity of a flexible parametric curve is still a pointwise constraint (unless one squares or exponentiates, of which more below), and two integral constraints remain: total mass one, and the martingale $\int e^x f_X(x) dx = 1$.

The quantile chart. Parametrize the quantile function $Q = F_X^{-1}$ on the rank $u \in (0, 1)$. One constraint: Q strictly increasing. And here the structure finally cooperates, because monotonicity — unlike convexity or positivity-of-a-difference — is destroyed by *one* derivative, not two.

The free chart. Let $q(u) = Q'(u)$ be the *quantile density*. Then Q increasing $\iff q > 0 \iff \log q$ is *any* real function. Modelling $\log q$ removes the last pointwise constraint; what remains are one integrability condition (so that the martingale integral converges) and one scalar normalization (to set its value to one) — and, remarkably, we will see in sections 4 and 7 that these cost a single inequality and a single additive shift. Table 2 tallies the trip, and fig. 2 shows the same

fitted slice in four of the charts: the constraints that are visibly at work in the first three panels are simply *absent* in the fourth.

Chart	Object	Constraints left over	Type
Price	$C(k)$	decreasing, convex, endpoints	pointwise, global
Volatility	$\sigma(k)$	$f \geq 0$ as a nonlinear ODE expression	implicit, global
Distribution	F_X	increasing	pointwise
Density	f_X	$f \geq 0$; mass = 1; $\mathbb{E}[e^X] = 1$	pointwise + 2 integrals
Quantile	$Q(u)$	increasing	pointwise
Log quantile density	$\log q(u)$	$A_R < 1$; one scalar shift	<i>one inequality</i>

Table 2: The shopping trip. Each chart describes the same space of mean-one laws; only the last reduces its boundary to a single scalar inequality, with the normalization absorbed by a shift rather than a constraint.

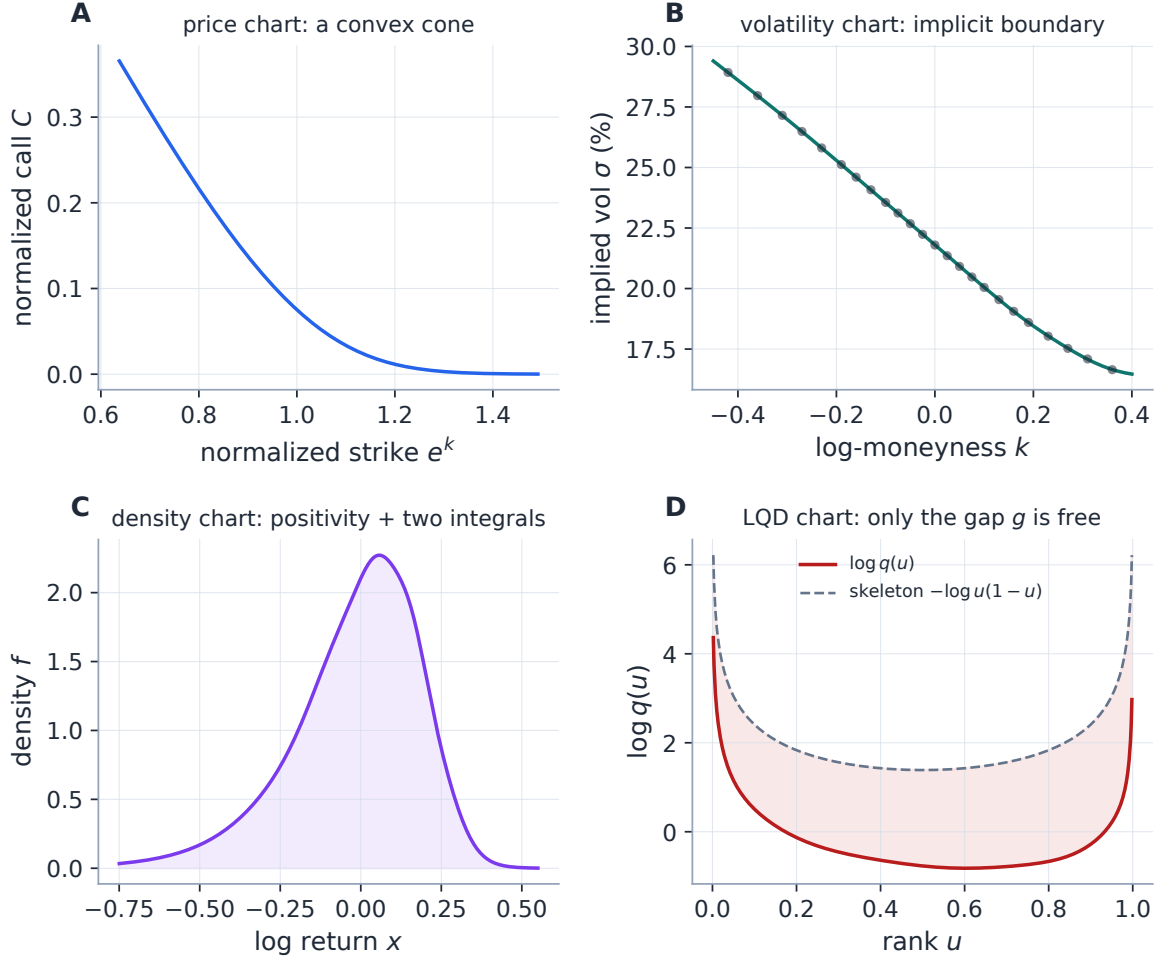


Figure 2: One fitted slice — the running SPX-like example — in four coordinate charts. A: the price chart, where no-arbitrage is a convex cone. B: the volatility chart, where it is an implicit nonlinear boundary (the quotes ride the fitted curve). C: the density chart, where it is positivity plus two integral constraints. D: the LQD chart of definition 2: the classical U of $\log q$, diving to $+\infty$ at both ends along the universal skeleton $-\log u - \log(1-u)$ (dashed). The skeleton is the same for every slice; the model's entire freedom is the shaded gap g between the two curves, and *any* smooth gap — shifted, bent, or oscillating — is another arbitrage-free slice, provided only that the single inequality $A_R < 1$ holds at the right edge. The fit's martingale check is $|\mathbb{E}[e^X] - 1| = 1.11 \times 10^{-16}$.

3.1 The model in one equation

It remains to choose a basis for $\log q$, and the choice is forced by two requirements: the tails must be *right by construction* (next section), and the body must be flexible, well-conditioned, and cheap. The LQD answer splits $\log q$ into a universal singular skeleton and a smooth part:

Definition 2 (The LQD slice). Fix an order $N \geq 4$ and coefficients $\theta = (L, R, a_2, \dots, a_N)$. The LQD quantile density is

$$q(u) = \frac{e^{g(u)}}{u(1-u)}, \quad g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u), \quad (5)$$

for $u \in (0, 1)$, with P_n the Legendre polynomials. The quantile is $Q(u) = \mu + \int_{1/2}^u q$, with the scalar μ fixed by the martingale condition $\mathbb{E}[e^X] = \int_0^1 e^{Q(u)} du = 1$.

Equivalently $\log q(u) = -\log u - \log(1-u) + g(u)$: the two logarithmic terms are the skeleton, the smooth g is everything else. The shipped default $N = 6$ carries seven parameters; the API accepts $4 \leq N \leq 16$, and higher N adds body detail without ever touching the no-arbitrage mechanism. Inverting $q = 1/f_X(Q)$ gives the density at the rank u :

$$f_X(Q(u)) = \frac{1}{q(u)} = u(1-u)e^{-g(u)} \geq 0 \quad \text{for every } \theta, \quad (6)$$

and this single line is the whole no-arbitrage mechanism. In code:

Listing 1: The LQD density is structurally non-negative — no constraint, no penalty (distilled from `models/lqd/quadrature.py`).

```
1 def density(u, g):          # g(u) = (1-u) L + u R + sum a_n P_n(1-2u)
2     return u * (1 - u) * np.exp(-g)    # >= 0 for ANY g: no butterfly
```

Proposition 1 (A single LQD slice is butterfly-free). *Suppose $q > 0$ on $(0, 1)$ and the martingale integral $\int_0^1 e^{Q(u)} du$ is finite. After the scalar shift of Q that makes this integral one, the call curve $C(k) = \int_{u_k}^1 (e^{Q(u)} - e^k) du$, with u_k the rank where $Q(u_k) = k$, is generated by a mean-one probability law on $Y > 0$: the slice is free of butterfly arbitrage, and put-call parity $C(k) - P(k) = 1 - e^k$ holds identically.*

Proof. Since $q > 0$, Q is strictly increasing and is the quantile of a genuine law for X ; then $Y = e^X > 0$ and the shift enforces $\mathbb{E}[Y] = 1$. The displayed $C(k)$ is just $\mathbb{E}[(Y - e^k)^+]$ written rank-by-rank (u is a uniform variable and $Y = e^{Q(u)}$), so monotonicity and convexity in strike follow from (4); the put integral $P(k) = \int_0^{u_k} (e^k - e^{Q(u)}) du$ subtracts to $C - P = \int_0^1 (e^Q - e^k) = 1 - e^k$. No positivity inequality is ever checked. $\square$

Heuristic

Why does the quantile side succeed where the density side struggled? Because exponentiating kills a *pointwise* constraint but not an *integral* one. On the density side, $f = e^\phi$ buys positivity but leaves two integral constraints ($\int f = 1$, $\int e^x f = 1$) coupling all parameters nonlinearly. On the quantile side, $q = e^\ell$ buys monotonicity, the mass constraint is *automatic* (every increasing Q on $(0, 1)$ is a quantile — ranks are born normalized), and the one remaining integral constraint is solved, not imposed: shifting Q by a constant multiplies $\mathbb{E}[e^X]$ by e^μ , so one scalar μ absorbs it exactly, leaving the shape of the law untouched. One chart, zero pointwise constraints, zero residual equality constraints.

Why this is the right trade. A direct-volatility model spends optimizer effort policing arbitrage; the LQD model spends none, because the feasible set *is* the set of laws. The price is two scalar nonlinearities — the shift μ (one logarithm of one integral, section 7) and the tail admissibility $A_R < 1$ (section 4) — both imposed once per residual evaluation, not at every quote.

4 Tails first: the endpoint skeleton

A basis for g should be chosen tails-first, because extrapolation is where smile models die. The far wings of a slice are constrained by a classical and completely model-free result, Lee's moment formula [2], and the two logarithmic terms in (5) are engineered so that LQD lands *inside* Lee's bounds automatically, with the wing slopes carried by two readable parameters. This section owns that story; the cross-model treatment of wings is Note 09's.

4.1 Endpoint scales are quantile slopes

The skeleton terms diverge at the ranks' endpoints, and the smooth part g decides *how fast*. Because $P_n(1) = 1$ and $P_n(-1) = (-1)^n$, the two endpoint values of g are

$$A_L := e^{g(0)} = \exp\left(L + \sum_{n \geq 2} a_n\right), \quad A_R := e^{g(1)} = \exp\left(R + \sum_{n \geq 2} (-1)^n a_n\right). \quad (7)$$

Their meaning is best read in the log-odds coordinate $z = \log \frac{u}{1-u}$ that will run the whole pricing machine. Since (writing $u(z)$ for the inverse map $u = 1/(1 + e^{-z})$, whose derivative is $u(1-u)$)

$$\frac{dQ}{dz} = q(u) u(1-u) = e^{g(u(z))}, \quad (8)$$

the quantile has *finite slope* e^g in z everywhere, and at the two ends

$$Q(z) = \mu + A_L z + O(1) \quad (z \rightarrow -\infty), \quad Q(z) = \mu + A_R z + O(1) \quad (z \rightarrow +\infty). \quad (9)$$

So A_L and A_R are the asymptotic speeds at which log-return quantiles recede per unit of log-odds: genuine tail *scales*, not diagnostics.

4.2 Exponential tails, the moment strip, and the one hard wall

Linear quantiles in log-odds mean exponential tails in return space. To see it, ask how much probability lives above the return level $x = Q(z)$: exactly the rank remainder $1 - u(z)$, which behaves like e^{-z} for large z . Inverting the asymptote $x \approx \mu + A_R z$ gives $z \approx (x - \mu)/A_R$, so

$$\mathbb{P}(X > x) \asymp e^{-x/A_R} \quad (x \rightarrow +\infty), \quad \mathbb{P}(X < x) \asymp e^{x/A_L} \quad (x \rightarrow -\infty): \quad (10)$$

each tail is exponential, with decay rate the reciprocal of its scale.

Moments now follow from a single change of variables, and it is worth doing once slowly because the whole admissibility story drops out. Any expectation is a rank integral, and in log-odds the rank measure carries an intrinsic tent-shaped weight:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u)) du = \int_{-\infty}^{\infty} h(Q(z)) \underbrace{u(z)(1-u(z))}_{\asymp e^{-|z|}} dz. \quad (11)$$

Take $h(x) = e^{rx}$. On the right, $e^{rQ(z)} \asymp e^{rA_R z}$ against the weight e^{-z} : the integrand behaves like $e^{(rA_R - 1)z}$, integrable at $+\infty$ iff $rA_R < 1$. On the left, $e^{rQ(z)} \asymp e^{rA_L z}$ against the weight e^{+z} : the integrand behaves like $e^{(rA_L + 1)z}$, integrable at $-\infty$ iff $rA_L > -1$. So $\mathbb{E}[e^{rX}]$ converges precisely on

the strip

$$-\frac{1}{A_L} < r < \frac{1}{A_R}, \quad (12)$$

each endpoint lost exactly when the exponential growth of e^{rQ} ties the exponential decay of the weight. In return moments — recalling $Y = e^X$ — the strip says that the last finite moments are

$$p_R^* = \sup\{p \geq 0 : \mathbb{E}[Y^{1+p}] < \infty\} = \frac{1}{A_R} - 1, \quad p_L^* = \sup\{p \geq 0 : \mathbb{E}[Y^{-p}] < \infty\} = \frac{1}{A_L}. \quad (13)$$

One point of the strip is load-bearing. The martingale integral $\mathbb{E}[e^X]$ is the case $r = 1$, and $r = 1$ lies inside the strip iff

$$A_R < 1 \quad (\text{finite forward}) \quad (14)$$

— the single wall promised in table 2. No left-hand condition exists because $r = 1 > -1/A_L$ always. The wall is not an artefact of the basis: any law whose right return tail is exponential with scale ≥ 1 has an infinite mean, whatever model writes it down.

Caution

$A_R < 1$ is a genuine hard constraint, not a regularization preference: a parameter vector with $A_R \geq 1$ assigns the forward an *infinite* mean, and its “prices” are meaningless. The calibrator enforces it twice — a hard rejection at $A_R \geq 1 - \varepsilon_R$ and a smooth softplus barrier that engages well before the wall (section 9) — so both the analytic and the finite-difference Jacobians stay smooth as the optimizer approaches the boundary. Fits that live near the wall should be treated as fragile even when admissible (section 14).

4.3 From moments to Lee slopes

Lee’s moment formula turns (13) into wing geometry. Writing $\beta_R = \limsup_{k \rightarrow +\infty} w(k)/k$ and $\beta_L = \limsup_{k \rightarrow -\infty} w(k)/|k|$ for the asymptotic slopes of total implied variance, Lee proved

$$\beta = \psi(p), \quad \psi(p) = 2 - 4(\sqrt{p^2 + p} - p) \in (0, 2], \quad (15)$$

with p the relevant critical exponent from (13). For LQD,

$$\beta_L = \psi\left(\frac{1}{A_L}\right), \quad \beta_R = \psi\left(\frac{1}{A_R} - 1\right), \quad A_R < 1. \quad (16)$$

The endpoint coefficients therefore do double duty: they regularize extrapolation *and* identify the moment budget and the Lee-consistent wing slopes. A calibration may fit L, R freely (subject only to the wall) or impose explicit wing priors by pinning A_L, A_R through (7). Figure 3 draws the two maps.

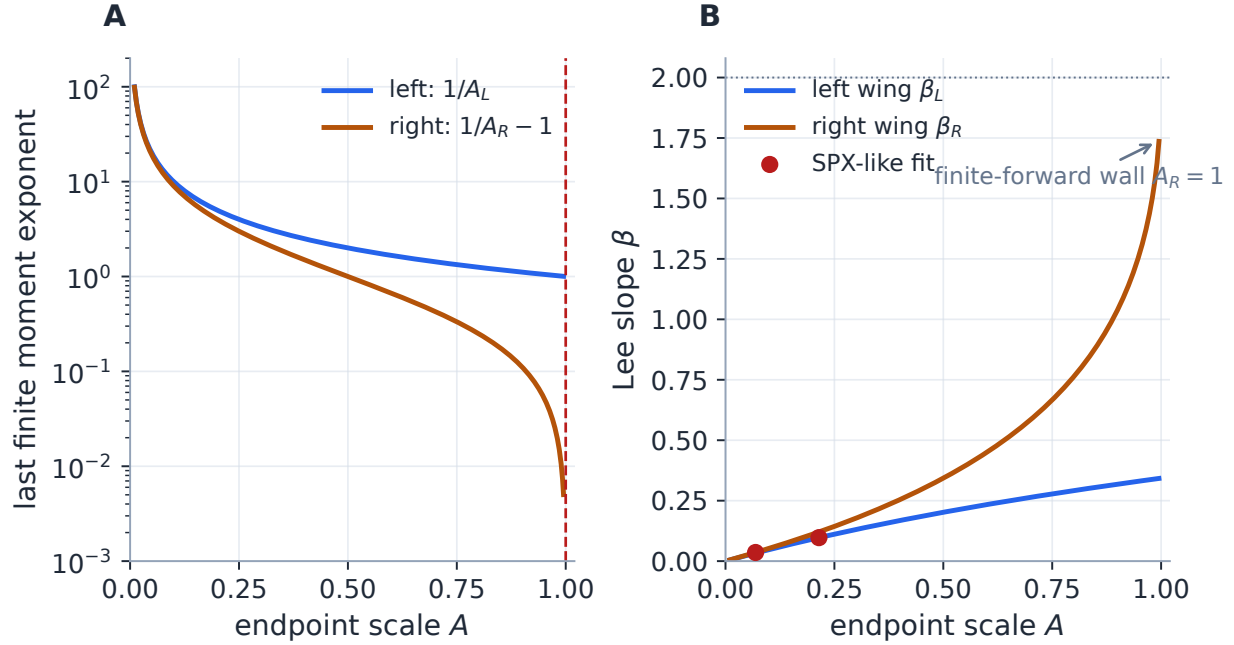


Figure 3: The tail dial. Panel A: the last finite moment exponent as a function of the endpoint scale, $1/A_L$ on the left tail and $1/A_R - 1$ on the right; the dashed line is the finite-forward wall $A_R = 1$, where the moment budget of the right tail is exhausted. Panel B: the Lee map ψ carries the same scales to asymptotic wing slopes, capped by the model-free ceiling $\beta = 2$; the dots mark an SPX-like production fit, sitting far below the ceiling — equity smiles use only a sliver of the admissible range.

Heuristic

The map ψ is increasing in the tail scale: a *heavier* tail (larger A , smaller critical exponent) makes a *steeper* variance wing. Be precise about “fat”: LQD return tails are *exponential* — asymptotically far heavier than the log-normal’s Gaussian log-return tails, though over traded strikes the two can look close. Equity index fits land at small A_L, A_R (our running example fits $A_L = 0.157473$, $A_R = 0.037786$), so their wings sit comfortably inside Lee’s bounds, and truncating the log-odds line at $|z| \leq 40$ costs nothing at double precision (section 7).

5 The Legendre body

With the tails owned by the skeleton, the body of g wants a basis that is smooth, global, well-conditioned, and orthogonal — deviations along different basis directions should not masquerade as one another. On $x = 1 - 2u \in (-1, 1)$ the Legendre polynomials satisfy

$$\int_0^1 P_m(1 - 2u) P_n(1 - 2u) du = \frac{\mathbf{1}_{m=n}}{2n + 1}, \quad (17)$$

and are generated by the stable three-term recursion

$$(n + 1)P_{n+1}(x) = (2n + 1)xP_n(x) - nP_{n-1}(x), \quad P_0 = 1, P_1 = x. \quad (18)$$

The expansion in (5) starts at $n = 2$ deliberately: degrees zero and one are already spanned by the endpoint terms, since

$$(1 - u)L + uR = \frac{L + R}{2} + \frac{L - R}{2}(1 - 2u). \quad (19)$$

So the $N = 6$ model spans exactly the polynomial space of $P_0, \dots, P_6$ while keeping the first two coordinates meaningful as *tail* handles rather than abstract polynomial coefficients: $\frac{L+R}{2}$ is an overall scale (roughly, the level of volatility) and $\frac{L-R}{2}$ a tail asymmetry (roughly, the skew).

What does a body mode actually *do* to a distribution? The mechanism is worth internalizing, because it is how every LQD fit works. Raising g near some rank u stretches the quantile spacing there ($q = e^g/u(1 - u)$ grows), and stretched spacing is *thinned probability*: the density $1/q$ falls on that stretch of returns while the displaced mass reappears wherever g was lowered. A body mode is therefore a mass-shuffling instruction written in rank coordinates, and its symmetry decides its trading meaning. $P_n(1 - 2u)$ is symmetric under $u \leftrightarrow 1 - u$ for even n and antisymmetric for odd n , so *even modes make symmetric reshapes, odd modes make skew-like ones*. Concretely, for the first three (shown in fig. 4 on a symmetric reference):

- $a_2 > 0$: P_2 is +1 at both endpoints and $-\frac{1}{2}$ at the median, so both tails stretch while the centre compresses — a sharper peak, fatter tails, a more convex smile. This is the butterfly mode.
- $a_3 > 0$: P_3 is antisymmetric, raising one tail's g and lowering the other's — mass slides from one side to the other, and the smile tilts. This is the risk-reversal mode.
- a_4 : one more oscillation of the same game, sculpting the *shoulders* between body and tail without moving the centre or the extreme tails much.

Higher modes continue the pattern with finer brushes. And because fig. 4 shows the same three perturbations in three charts at once, it is also a small test of the reader's new fluency: cover two panels and predict the third.

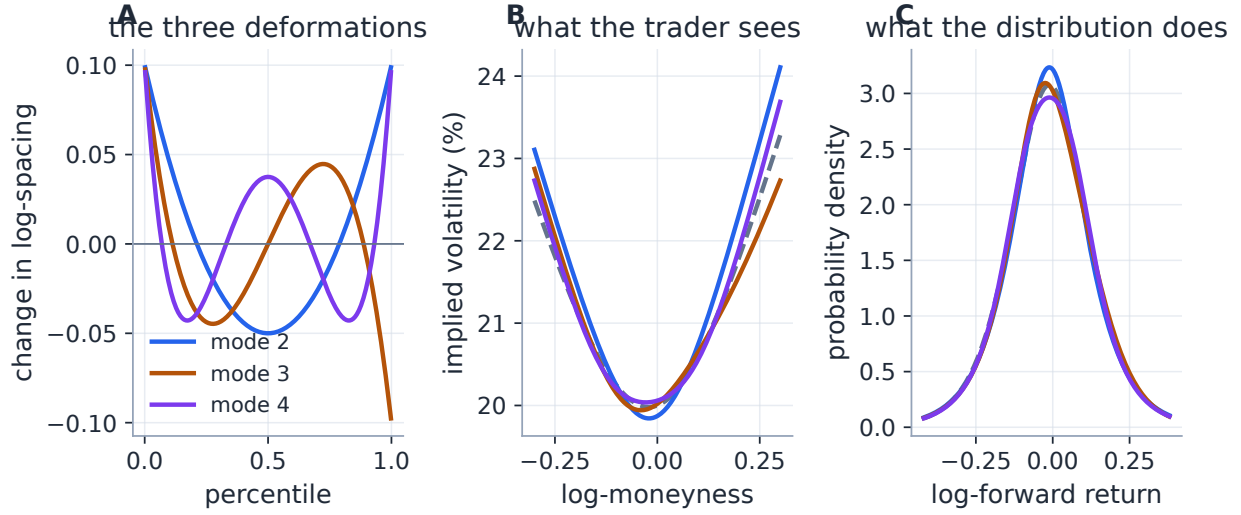


Figure 4: One production basis coefficient at a time: a_2 , a_3 or a_4 set to 0.10 on the symmetric logistic reference of section 6, everything else untouched. Panel A: the perturbation of g — a pure Legendre mode in the rank, even ones symmetric, odd ones not. Panel B: what the trader sees — mode 2 bends the smile symmetrically (butterfly), mode 3 tilts it (risk reversal), mode 4 works the shoulders. Panel C: what the distribution does — mass leaves the stretched ranks and lands on the compressed ones, and every deformed density is non-negative without anyone checking. The moral: *body modes shuffle mass; endpoint terms own the tails*.

6 An exactly solvable slice

Every lecture needs its harmonic oscillator. Set all body modes to zero and the two endpoint coefficients equal:

$$a_n = 0, \quad L = R = \log s \quad \implies \quad g \equiv \log s, \quad \frac{dQ}{dz} = s. \quad (20)$$

The quantile is exactly linear in log-odds, $Q(z) = \mu + sz$; that is, X is a *logistic* random variable with scale s : $X = \mu + sZ$ with $Z = \log \frac{U}{1-U}$ for a uniform rank U .

Everything about this slice flows from one integral. The moment generating function of Z is a rank integral we can actually do — substituting $e^{tZ} = \left(\frac{u}{1-u}\right)^t$ turns it into Euler's beta integral:

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u}\right)^t du = B(1+t, 1-t) = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \quad |t| < 1. \quad (21)$$

Note, in passing, that the formula itself confirms section 4: it exists exactly on the strip $|t| < 1$, as it must for a law with $A_L = A_R = 1$ per unit scale. Two facts fall out of (21) at once:

- **The martingale shift, exactly.** Read it at $t = s$: $\mathbb{E}[e^{sZ}] = \pi s / \sin(\pi s)$, so the shift that makes e^X mean-one is $\mu = -\log(\pi s / \sin(\pi s))$ in closed form — the only slice in the family whose μ needs no quadrature.
- **The variance, and hence the initializer.** Expand its logarithm at $t = 0$: since $\sin(\pi t) =$

$$\pi t(1 - \pi^2 t^2/6 + \dots),$$

$$\log \mathbb{E}[e^{tZ}] = \frac{\pi^2 t^2}{6} + O(t^4) \quad \implies \quad \text{Var}(Z) = 2 \cdot \frac{\pi^2}{6} = \frac{\pi^2}{3},$$

the second cumulant being twice the quadratic coefficient. So $\text{Var}(X) = \pi^2 s^2/3$, and matching a target ATM total variance w_0 gives the seed $s \approx \sqrt{3w_0}/\pi$.

The remaining diagnostics are immediate: $A_L = A_R = s$, so the slice is admissible iff $s < 1$ (a 55% vol at one year is $s \approx 0.30$, nowhere near the wall), and the Lee slopes are $\beta_L = \psi(1/s)$ and $\beta_R = \psi(1/s - 1)$ — slightly asymmetric wings even for this symmetric law, because the martingale shift tilts the smile.

This two-line model is precisely the production cold-start initializer — the `logistic_init` routine of `models/lqd/calibrate.py`: match s to the ATM-interpolated quote variance, then let the optimizer bend $L \neq R$ (tail asymmetry) and the a_n (body).

Figure 5 walks the whole pipeline on a solved instance — production root-solving the scale so that the ATM volatility is *exactly* 20% at $\tau = 0.50$ — and it repays reading one panel at a time. Panel A is the coordinate itself: the log-odds line holds every percentile of an unbounded law at finite coordinates, with equal steps in z meaning equal *multiplications* of the odds (the 1st and 99th percentiles sit at $z = \mp 4.6$, not at infinity). Panel B is the law: in rank coordinates the straight line $Q = \mu + sz$ becomes the gentle S-curve shown, and one detail deserves a pause — the forward ($k = 0$, marked) sits at the 53.3436% rank, *above* the median. That is the martingale shift at work: for e^X to average to one, the convexity of the exponential demands a median return slightly below zero ($Q(\frac{1}{2}) = \mu = -0.01088288$), so more than half the outcomes finish below the forward. Jensen's inequality, drawn. Panel C is the market's view of the same object: a symmetric, gently convex smile pinned at 20.00% — convex because exponential tails are heavier than the log-normal's, so far strikes carry more relative value, which the Black formula can only express as higher implied volatility. The example box then prices one option on this slice by hand.

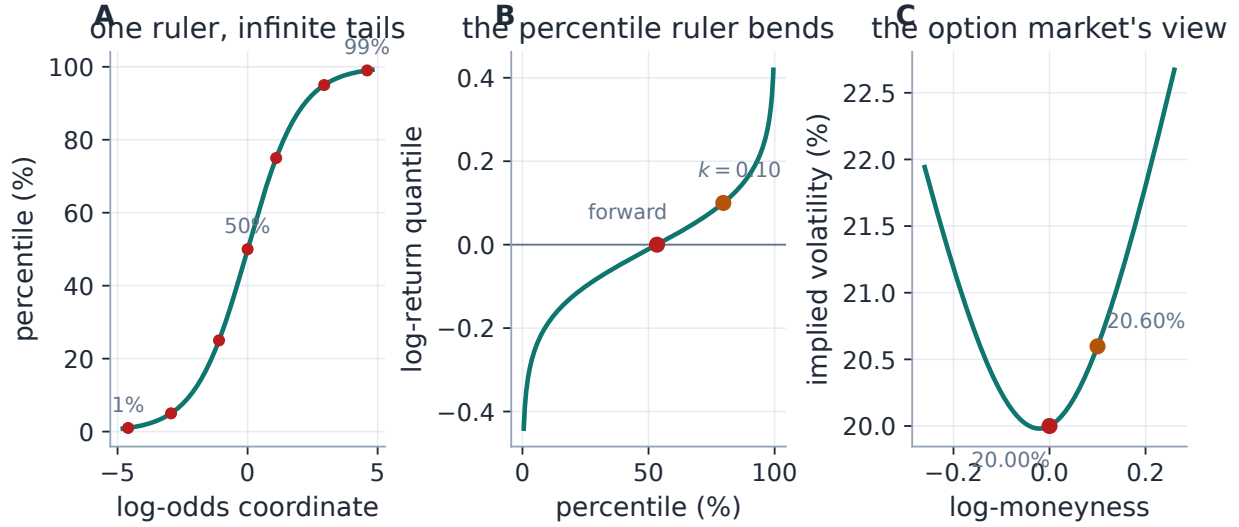


Figure 5: The exactly solvable slice, solved to an ATM volatility of exactly 20% at $\tau = 0.50$ (production root-solve: $s = 0.08125025$). A: the log-odds ruler — every percentile at a finite coordinate. B: the quantile in rank coordinates; the forward’s rank is $53.3436\% > 50\%$ because the martingale shift pushes the median return down to $\mu = -0.01088288$ (Jensen’s inequality, drawn). C: the resulting smile — 20.00% at the money by construction, convex because exponential tails out-fatten the log-normal.

A trade ticket priced by hand

On the solved slice of fig. 5, price the $k = 0.10$ call ($\approx 10.5\%$ out of the money). The strike’s rank is $u_k = 0.796524$: the option finishes in the money with probability $1 - u_k \approx 20.3\%$. The two ledgers of section 7 read $G(z_k) = 0.24722094$ (the share of the mean carried by outcomes above the strike) and $e^k(1 - u_k) = 0.22487593$ (the strike paid on that event), so

$$C(0.10) = G - e^k(1 - u_k) = 0.02234501 \quad \implies \quad \sigma_{\text{imp}} = 20.5977\%,$$

about 0.6 vol points above the ATM level: the first taste of a smile. Production reproduces every number in this box to the displayed digits (`gen_lqd_fresh.py`); the martingale check on the slice returns $\mathbb{E}[e^X] - 1 = 0$ to machine precision.

7 From coefficients to prices: one cumulative integral

The whole pricing pipeline is the arrow chain

$$\theta \longrightarrow g \longrightarrow Q \longrightarrow \mu \longrightarrow C(k) \longrightarrow \sigma(k),$$

and this section makes each arrow one integral (or one root-solve) on a single shared grid.

7.1 The log-odds machine

The skeleton’s endpoint singularities are a feature for tails and a nuisance for quadrature — so we integrate in the coordinate that removes them. Under $z = \log \frac{u}{1-u}$, by (8), the quantile is the

integral of a smooth positive function on the whole line:

$$Q(z) = \mu + \int_0^z e^{g(u(s))} ds, \quad (22)$$

anchored at the median node $z = 0$. The shift is fixed by the martingale condition, which in the same coordinate reads (using $du = u(1 - u) dz$)

$$\mu = -\log\left(\int_{-\infty}^{\infty} e^{Q(z)-\mu} u(z)(1 - u(z)) dz\right), \quad (23)$$

one logarithm of one integral. This is the *only* normalization in the model; being an additive shift of Q it changes neither q , nor the tails, nor positivity — it slides the whole law along the return axis until the forward is its mean.

Pricing needs one more cumulative object. Define the *upper share*

$$G(z) = \int_z^{\infty} e^{Q(s)} u(s)(1 - u(s)) ds = \mathbb{E}[Y \mathbf{1}_{\{Y > e^{Q(z)}\}}] : \quad (24)$$

the share of the (unit) mean carried by outcomes above the rank $u(z)$. It falls from $G(-\infty) = 1$ to $G(+\infty) = 0$, and it prices every call at once: if z_k solves $Q(z_k) = k$, then, splitting the call payoff into asset and cash legs,

$$C(k) = G(z_k) - e^k (1 - u(z_k)). \quad (25)$$

The put is $P(k) = e^k u(z_k) - (1 - G(z_k))$, and parity is immediate. Equation (25) is worth staring at: an option price is the difference of two ledgers, both monotone, both already on the grid. The same G will hand us the analytic Jacobian in section 10 and the calendar ordering in section 11 — one object, three jobs.

7.2 What one build of a slice actually is

It is worth being concrete here, because the entire calibrator stands on this one loop being cheap and smooth. Given a trial θ , production does five things, in order, on a fixed symmetric grid $z_j \in [-Z, Z]$:

1. evaluate g at the (cached) ranks u_j — one matrix-vector product, since g is linear in θ ;
2. integrate e^g cumulatively (fourth-order Simpson) to get the unshifted quantile, anchored at the median node $z = 0$;
3. compute the martingale integral, take one logarithm, and shift: $Q += \mu$;
4. integrate $e^Q u(1 - u)$ cumulatively *from the right* to get the upper share G ;
5. stop. Every strike, at any later moment, is priced from (25) by interpolating these two curves and solving one root.

Nothing here is iterative, nothing can fail for an admissible θ , and the cost is a few dense passes over one array.

Two details of this recipe deserve their “why”. The first is the truncation at $|z| = Z$ (production: $Z = 40$). Beyond the grid the integrands of steps 3 and 4 are, by section 4, essentially exact

exponentials — so instead of praying the tail is negligible, production *sums it in closed form*, appending the geometric-series corrections

$$\int_Z^\infty e^{Qu}(1-u) dz \approx \frac{e^{Q(Z)-Z}}{1-A_R}, \quad \int_{-\infty}^{-Z} e^{Qu}(1-u) dz \approx \frac{e^{Q(-Z)-Z}}{1+A_L}. \quad (26)$$

(Notice the right-hand denominator displaying the $A_R = 1$ wall once more.) For equity tail scales the corrections themselves are far below double-precision noise at $Z = 40$; they are kept because they make the build exact-in-the-limit rather than merely truncated.

The second detail is how the two curves are read *between* nodes. The construction hands us the exact nodal derivatives for free — $dQ/dz = e^g$ from (8) and $dG/dz = -e^Qu(1-u)$ from (24) — and it would be wasteful not to use them: production stores both and interpolates off-grid with *cubic Hermite* polynomials rather than linearly. The payoff is threefold: priced curves are smooth enough for clean finite-difference Greeks; the strike root $Q(z_k) = k$ can be polished by Newton on the interpolant to machine precision (which is what will make the ATM handles of section 8 *exact* rather than grid-limited); and because Hermite interpolation returns the nodal value exactly at a node, a constraint expressed at grid z -values — the calendar floor of section 11 — matches its node-indexed form bit for bit, whatever grid the optimizer runs on.

Performance

The grid $(z_j, u_j, u_j(1-u_j))$ and the Legendre matrix $P_n(1-2u_j)$ depend only on (Z, n_{pts}, N) — never on θ . A calibration builds the slice many hundreds of times, so these arrays are LRU-cached and returned read-only; only the parameter combination g is re-formed per call, which alone made `build_slice` about $1.7\times$ faster. Optimization iterates on a coarser grid of 2001 nodes (Simpson error already orders of magnitude below the fit budget) and the accepted slice is rebuilt once at the full 8001 nodes for reporting, density, var-swap and calendar diagnostics. Converged parameters agree with a full-grid fit to $\sim 10^{-6}$. See appendix B.

7.3 Back to the quoted smile

The market speaks implied volatility, so the last arrow inverts Black. In normalized total-variance units,

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \quad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \quad (27)$$

and the model's total implied variance solves $B(k, w(k)) = C(k)$ — a scalar root per strike, monotone in w , safe for Newton.

Calibration, however, deliberately does *not* fit in volatility space. Given a quote σ_i at k_i , with $w_i = \sigma_i^2 \tau$, the residual is the *vega-normalized price error*

$$r_i(\theta) = \frac{C(k_i; \theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta}, \quad \partial_\sigma B = \varphi(d_+) \sqrt{\tau}, \quad (28)$$

with a small floor η against the vanishing far-wing vega (appendix A). Working in price space keeps every iterate a valid law (no implied-vol inversion inside the loop, no inversion failures on wild iterates); dividing by vega undoes the price-to-vol scale so the optimizer still sees, to first order, a volatility error. The error *ledger* the note reports — always in vol basis points — is computed after convergence by honest Black inversion.

8 Three exact handles, and a chart built around them

8.1 Level, skew, curvature in closed form

Traders want the leading coordinates of a smile model to be the *actual* ATM volatility, skew and curvature — not proxies, not finite differences of a plotted curve. The LQD slice delivers all three exactly, and the derivation is three short steps: read two numbers off the law, pin the ATM level, then push everything through the Black formula with the chain rule. Recall the convention of table 1: primes are k -derivatives of one-variable curves (C, w, σ), subscripted ∂ 's are partials of the two-variable $B(k, w)$; everything below is evaluated at the money.

Step 1: two numbers from the law. Differentiate the rank-form call $C(k) = \int_{u_k}^1 (e^{Q(u)} - e^k) du$ in k . The moving boundary u_k contributes nothing — the integrand vanishes exactly there, an envelope fact that returns in force in section 10 — so only the $-e^k$ inside survives: $C'(k) = -e^k(1 - u_k)$. Differentiating once more brings in the density, since $du_k/dk = f_X(k)$. At $k = 0$, writing u_0 for the forward's rank ($Q(u_0) = 0$) and $f_0 = u_0(1 - u_0)e^{-g(u_0)}$ for the ATM density (6),

$$C'(0) = -(1 - u_0), \quad C''(0) = f_0 - (1 - u_0). \quad (29)$$

Step 2: the ATM level. At $k = 0$ the Black formula collapses, and the ATM total variance w_0 is pinned by one monotone scalar equation,

$$B(0, w_0) = 2\Phi\left(\frac{\sqrt{w_0}}{2}\right) - 1 = C(0). \quad (30)$$

Step 3: the chain rule through Black. The model's smile is *defined* by the identity $B(k, w(k)) = C(k)$. Differentiating it once and twice at $k = 0$,

$$\partial_k B + \partial_w B w' = C', \quad \partial_{kk} B + 2\partial_{kw} B w' + \partial_{ww} B (w')^2 + \partial_w B w'' = C'',$$

and solving for the two unknowns,

$$w'(0) = \frac{C'(0) - \partial_k B}{\partial_w B}, \quad w''(0) = \frac{C''(0) - \partial_{kk} B - 2\partial_{kw} B w'(0) - \partial_{ww} B (w'(0))^2}{\partial_w B}. \quad (31)$$

The five ATM Black partials are classical; with the shorthands $a = \sqrt{w_0}/2$, $\Phi_a = \Phi(a)$, $\varphi_a = \varphi(a)$,

$$\begin{aligned} \partial_k B &= -(1 - \Phi_a), & \partial_w B &= \frac{\varphi_a}{2\sqrt{w_0}}, & \partial_{kk} B &= -(1 - \Phi_a) + \frac{\varphi_a}{\sqrt{w_0}}, \\ \partial_{kw} B &= \frac{\varphi_a}{4\sqrt{w_0}}, & \partial_{ww} B &= -\frac{\varphi_a}{4w_0^{3/2}} - \frac{\varphi_a}{16\sqrt{w_0}}. \end{aligned} \quad (32)$$

Finally, translate total variance into volatility: $\sigma(k) = \sqrt{w(k)/\tau}$, so one more application of the same chain rule yields the handles,

$$\sigma_0 = \sigma(0) = \sqrt{\frac{w_0}{\tau}}, \quad s_0 = \sigma'(0) = \frac{w'(0)}{2\sqrt{\tau w_0}}, \quad \kappa_0 = \sigma''(0) = \frac{w''(0)}{2\sqrt{\tau w_0}} - \frac{w'(0)^2}{4\sqrt{\tau} w_0^{3/2}}. \quad (33)$$

Every quantity along the route is closed-form or one scalar root, so the handles are exact for every LQD slice: the level in volatility units (annualized on the active clock τ), the skew in vol per unit

log-moneyness, the curvature in vol per unit log-moneyness squared.

Remark 1 (One triple, two unit systems). Production carries this information in two *equivalent* coordinate systems: the graph-facing carrier $(\sigma_0, s_0, \kappa_0)$ that Note 14 propagates, and the internal chart's $H = (w_0, s_0, \kappa_0)$ with the ATM total variance first (`models/lqd/ortho.py`). Since $w_0 = \sigma_0^2 \tau$, the two are a smooth bijection — no information is at stake, only the *units* of the first coordinate. The reason to be pedantic is that a volatility handed where a total variance is expected fails silently; so the note fixes the convention once — H always means the total-variance triple — and the graph adapter converts.

Heuristic

Equation (29) is the distribution speaking directly. $-C'(0)$ is the probability of finishing above the forward — the ATM digital — and the skew in (31) is, at heart, the mismatch between that digital and the $\Phi(d_-)$ the Black model would assign it. The curvature adds one more distributional fact, the ATM density f_0 . Level, digital, density: three numbers of the law, three handles of the smile.

8.2 The ATM-orthogonal chart

The handles earn their keep when they stop being diagnostics and become *coordinates*: axes of parameter space that a person can hold. What we want is a chart around the current fit whose first three axes *are* level, skew and curvature, and whose remaining axes deliberately do nothing to them. Building it amounts to answering two questions, and each answer is a subspace.

Fix the reference θ^* (in practice, the freshly fitted slice), write $H(\theta) = (w_0, s_0, \kappa_0) \in \mathbb{R}^3$ for the handle map, and let $J = \partial H / \partial \theta \in \mathbb{R}^{3 \times d}$ be its Jacobian at θ^* — computed by central differences, which on this deterministic, smooth quadrature is accurate to $\sim 10^{-9}$.

Question 1: what is the cheapest way to move the handles? A perturbation $\delta\theta$ moves them by $J \delta\theta$ to first order, so a prescribed handle move δH means three equations in d unknowns — underdetermined, many solutions. Take the shortest one:

$$\delta\theta = U \delta H, \quad U = J^\top (J J^\top)^{-1}, \quad (34)$$

the least-norm right inverse of J , so that $JU = I_3$. The three columns of U are the *primary directions*: moving along them changes the handles one-for-one, in handle units, with no wasted motion in parameter space.

Question 2: which moves leave the handles alone? To first order, exactly the kernel of J — a $(d - 3)$ -dimensional space. Collect an orthonormal basis of it as the columns of V ; these are the *shape directions*.

Gluing the two answers together gives the chart:

$$\theta = \theta^* + U\alpha + V\xi, \quad (35)$$

with three coordinates α that read directly as handle moves — $H \approx H(\theta^*) + \alpha$ — and $d - 3$ coordinates ξ that reshape wings, shoulders and event convexity while the handles stand still.

First order is not forever, so production closes the gap. To hit a target H^{target} *exactly*, it solves the three-dimensional system $H(\theta^* + U\alpha + V\xi) = H^{\text{target}}$ for α by Newton — and here the construction pays for itself twice, because $JU = I_3$ makes the system's Jacobian the identity at the

reference: the natural first guess $\alpha = H^{\text{target}} - H(\theta^*)$ is already excellent, and the preconditioner rarely needs refreshing. This *retarget* is what lets a trader drag ATM level, skew or curvature while the optimizer keeps the shape modes for everything else; fig. 6 shows one move of each kind on the running example.

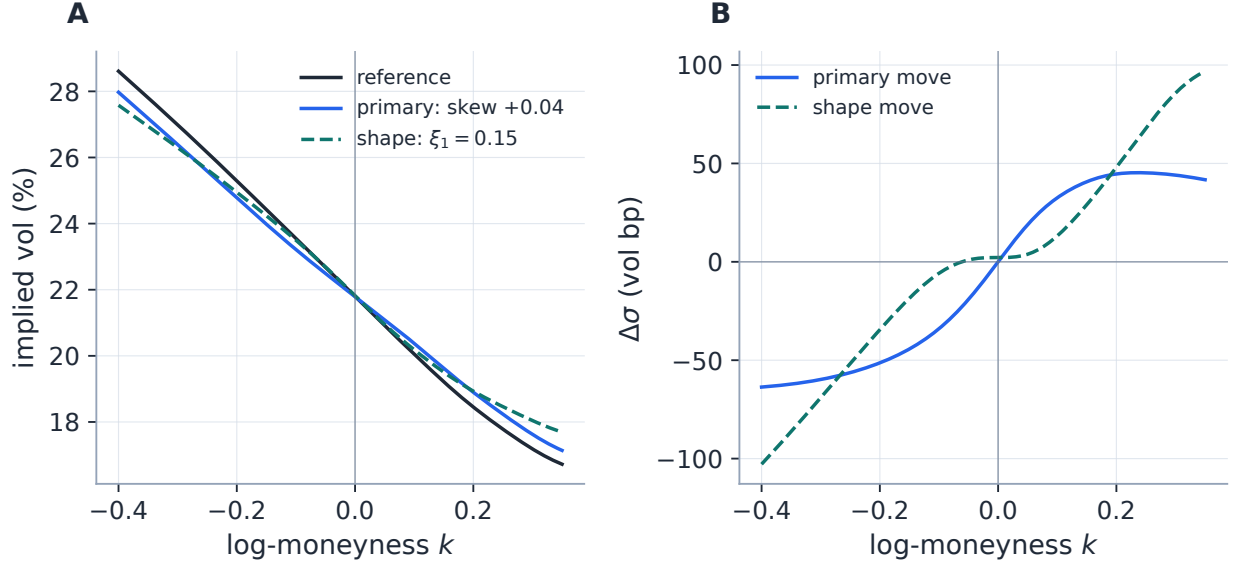


Figure 6: The chart in action on the running SPX-like fit ($d = 10$, so 7 shape directions). Panel A: the reference smile; a *primary* move retargeting the skew by +0.04 exactly (achieved to 6.00×10^{-14} by the Newton solve) with level and curvature held; and a *shape* move along the first kernel direction. Panel B: the same two moves as differences. The primary move tilts the smile through the money; the shape move relocates over 103 vol bp of wing while drifting the ATM level by only 2.17 vol bp, the skew by 7.88×10^{-5} and the curvature by 6.27×10^{-3} — first-order invariance visible to the eye as a flat spot at the money.

Caution

The chart is *local*: U and V are frozen at θ^* , so the handle-invariance of shape directions degrades quadratically with the excursion (the residual drift in fig. 6 is exactly that second order), and the retarget Newton solve can fail — or converge to an *infeasible* point — when the requested handles push the slice toward the $A_R = 1$ wall (a large curvature demand on a thin-tailed reference is the classic case). Callers must treat retargeting as an operation that can fail near the wall, not as a total map on handle space.

9 Calibrating one expiry

We have spent eight sections building a space in which every point is an arbitrage-free slice. Choosing the *right* point is now almost an anticlimax — which was the design goal all along. This section states the objective, the one place it must brake (the wall), where fits start from, and how the shipped objective is actually assembled.

9.1 The core objective

Given quotes (k_i, σ_i) at one expiry, the calibrator solves the bound-free nonlinear least-squares problem (`scipy trf`)

$$\min_{\theta} \sum_i r_i(\theta)^2 + \lambda \sum_{n \geq 4} n^{2r} a_n^2, \quad (36)$$

with r_i the vega-normalized residuals (28). (An optional per-quote weighting — Note 07’s `weightScheme` — multiplies the residuals; it changes nothing below and we suppress it.) The high-order ridge $\lambda n^{2r} a_n^2$ is optional; it suppresses oscillation in sparse markets while deliberately leaving the first body modes a_2, a_3 — the workhorses of curvature and asymmetry — untouched. Note what is *absent*: there is no density-positivity term anywhere, because positivity is structural (proposition 1). The one admissibility condition is the wall (14), enforced as $A_R < 1 - \varepsilon_R$ with a tiny buffer, plus the smooth barrier below.

9.2 The soft turn before the hard wall

To keep the residual smooth as the optimizer approaches $A_R = 1$, the stacked residual carries a softplus barrier row

$$b(\theta) = \log(1 + e^{\gamma(A_R - c)}), \quad (37)$$

with centre $c = 0.90$ and steepness $\gamma = 50.0$ (both `FitSettings` coefficients; evaluated as `logaddexp(0, .)`, so a wild trial θ cannot overflow it). The barrier is negligible for $A_R \ll c$ and rises steeply near the wall, turning the fit around *before* the hard rejection; if a trial θ does cross $A_R \geq 1$, the price block is replaced by a large penalty increasing smoothly in A_R , so the trust region still sees a gradient pointing back to feasibility.

9.3 Initialization and warm starts

The only cold-start initializer in production is the exactly solvable slice of section 6: $a_n = 0$, $L = R = \log s$ with $s = \sqrt{3w_0}/\pi$ from the variance match, seeded from the ATM-interpolated quote variance. Sequential workflows do better than cold: the background Calibrate sweep *warm-starts* each expiry from the previous, shorter-dated expiry’s freshly fitted parameters (adjacent maturities have nearly the same smile, so `trf` lands close to the optimum), and the service can also run a *data-only prepass* — a persistence-free fit used as the measurement stage of the prior and filter machinery (Notes 13 and 15). Wing-aware and quantile-projection initializers are roadmap items, not shipped code.

9.4 The full residual stack

The two-term objective (36) is the lecture version. The shipped objective is a *stack* of residual blocks, and table 3 is the map of the whole territory: the data block and ridge we have met, the calendar slack of section 11, the barrier, and then a run of optional blocks — var-swap targets, priors, filter terms — each of which belongs to its own note in the series and simply plugs extra rows into the same least-squares problem. Two properties of the assembly are worth carrying away. Every optional block is byte-identical-to-absent when unused (test-locked), so switching a feature off really is off. And the Jacobian column matters operationally: the analytic route of section 10 covers the first five blocks, while activating *any* var-swap or prior block switches the whole fit to `scipy`’s finite-difference Jacobian — correct, merely not accelerated.

Residual block		Active when	Units / scaling	Jacobian
Mid data block		<code>fitMode=mid</code> (default)	vega-normalized price ($\approx$ vol)	analytic
Band data block (hinge + mid anchor; Note 07)		<code>fitMode bidask/haicut</code>	vega-normalized price	analytic
High-order ridge		<code>regLambda > 0</code> (rows $n \geq 4$)	coefficient units	analytic
Calendar slack (Note 10)		<code>enforceCalendar</code> + a nearer fitted expiry	upper share (normalized price)	analytic
A_R soft barrier		always (one row)	dimensionless	analytic
Market (Note 08)	var-swap	<code>varSwapEnabled</code> + an active quote	vol	FD fallback
Prior (Note 13)	strike-gap anchor	<code>strike_gap</code> / <code>hybrid</code> active prior	vega-normalized price	FD fallback
Prior (Note 13)	var-swap companion	alongside an active prior	vol	FD fallback
Operator prior (Note 13)		<code>quote_operator</code> / <code>hybrid</code> gated per operator	operator units (vol)	FD fallback
Active-filter (Note 15)	MAP	filter mode <code>active</code> (rides the operator block, ungated)	handle units	FD fallback

Table 3: The LQD residual stack as production assembles it (`models/lqd/calibrate.py`; prior/filter targets are resolved in `api/service.py`). “FD fallback” means the presence of that block switches the entire fit to the finite-difference Jacobian.

9.5 The native var-swap route

The log-contract identity gives the var-swap proxy in one line of the model’s own coordinates:

$$w_{vs} = -2 \mathbb{E}[X] = -2 \int_{-\infty}^{\infty} Q(z) u(z) (1 - u(z)) dz, \quad (38)$$

a single quadrature on the grid the slice already owns, with integrand decaying like $z e^{-|z|}$ (`quadrature.py::var_swap_strike`). Two qualifications keep the claim honest. First, this native route is dramatically cheaper than the generic strike replication of Note 08 — which would re-solve implied variance on a strike grid at every Jacobian column — but activating the var-swap block also forfeits the analytic Jacobian (table 3), so a var-swap fit is not literally as cheap as a mid fit. Second, the identity prices the *log contract*: its correspondence with a realized-variance swap assumes continuous monitoring of a continuous (diffusive) path; jumps and discrete sampling take the usual corrections, outside this model’s scope.

10 The derivative that comes free

The dominant cost of (36) used to be the Jacobian: `scipy` rebuilt the entire quadrature pipeline $P + 1$ times per iteration to finite-difference it, with $P = N + 1$ the parameter count. The Vol-Fitter instead propagates the analytic Jacobian of the core residual stack — data block (mid or band), ridge, calendar slack, barrier — in *one* quadrature pass. The key is a cancellation identity, and it is the cleanest piece of mathematics in the implementation.

10.1 The cancellation identity

The subtlety is that the priced call (25) depends on θ twice: explicitly through the curves Q and G , and implicitly through the strike root $z_k(\theta)$ solving $Q(z_k; \theta) = k$. A naive gradient must differentiate the root. Differentiate the pricing identity and watch:

$$\frac{dC}{d\theta} = \underbrace{\frac{\partial G}{\partial \theta}}_{\text{explicit}} \Big|_{z=z_k} + \left[G'(z_k) + e^k u'(z_k) \right] \frac{dz_k}{d\theta}. \quad (39)$$

Now $G'(z) = -e^{Q(z)}u(1-u)$ from (24), while at the root $e^{Q(z_k)} = e^k$ and $u'(z_k) = u_k(1-u_k)$. The bracket is

$$-e^k u_k(1-u_k) + e^k u_k(1-u_k) = 0. \quad (40)$$

Theorem 1 (Implicit-root cancellation). *The implicit dependence of the strike root on the parameters drops out of the price gradient:*

$$\frac{dC}{d\theta} = \frac{\partial G}{\partial \theta} \Big|_{z=z_k}. \quad (41)$$

Remark 2. This is an envelope argument in disguise. $C(k)$ is an integral whose lower limit z_k sits exactly where the integrand of the combined asset-and-cash ledger vanishes — the payoff ($e^Q - e^k$) is zero at its own exercise boundary — so moving the boundary is free, to first order. The same mechanism makes the American exercise boundary drop out of American option Greeks.

So $dC/d\theta$ is obtained by evaluating the *nodal* parameter sensitivities of the upper share at the fixed point z_k — by the very Hermite interpolation used to price, fed $\partial G/\partial \theta_j$ and $\partial G'/\partial \theta_j$ instead of the nodal values.

10.2 Nodal sensitivities in one pass

With the root out of the way, there is nothing left to invent: we simply differentiate the five-step recipe of section 7.2, step by step, and every step stays a cumulative integral. Two structural facts make it painless. First, g is *affine* in θ , with constant basis $\phi_j = \partial g/\partial \theta_j$ — namely $\phi_L = 1 - u$, $\phi_R = u$, $\phi_{a_n} = P_n(1 - 2u)$ — so differentiating anything of the form e^g just multiplies it by a fixed, cached array. Second, integration and $\partial/\partial \theta_j$ commute, so a derivative of a cumulative integral is the cumulative integral of the derivative.

Walk the recipe. Step 2 (the unshifted quantile $\bar{Q} = Q - \mu$) differentiates under the integral sign,

$$\frac{\partial \bar{Q}(z)}{\partial \theta_j} = \int_0^z e^g \phi_j ds, \quad (42)$$

one more cumulative Simpson pass per parameter. Step 3 (the shift $\mu = -\log M$, with M the martingale integral) is a logarithm, so

$$\frac{\partial \mu}{\partial \theta_j} = -\frac{1}{M} \int e^{\bar{Q}} \frac{\partial \bar{Q}}{\partial \theta_j} u(1-u) dz + (\text{tail-correction terms}), \quad (43)$$

where the corrections (26) contribute through $\bar{Q}(\pm Z)$ and through A_L, A_R — whose own gradients cost nothing, since (7) gives $\partial A_L/\partial \theta = A_L(1, 0, 1, 1, \dots)$ and $\partial A_R/\partial \theta = A_R(0, 1, 1, -1, \dots)$ with signs $(-1)^n$. Step 4 (the share) is another cumulative pass: with $\partial Q = \partial \mu + \partial \bar{Q}$ in hand, integrate

$\partial(e^Q u(1-u))/\partial\theta_j$ from the right, add its right-tail correction, and note that the nodal slope sensitivity $\partial G'/\partial\theta_j$ is the same integrand negated — exactly the pair the Hermite interpolant wants. The non-price rows of the stack then differentiate in one line each: the band hinge multiplies $dC/d\theta$ by the violation sign, the ridge is diagonal on the a -columns, the calendar slack reuses the share sensitivity at the constraint nodes, and the barrier row is $\exp(\gamma(A_R - c)) \gamma \partial A_R/\partial\theta$. One quadrature pass, all P columns.

One honesty note on exactness. Theorem 1 is exact for the *continuous* objects. The discretization interpolates Q and G as two separate Hermite interpolants, so between nodes the interpolated price is not algebraically the integral of the interpolated quantile, and the implemented gradient inherits that (tiny) mismatch. The operational claim is therefore *benchmarked numerical agreement*, not algebraic identity — and it is benchmarked twice over, in fig. 7.

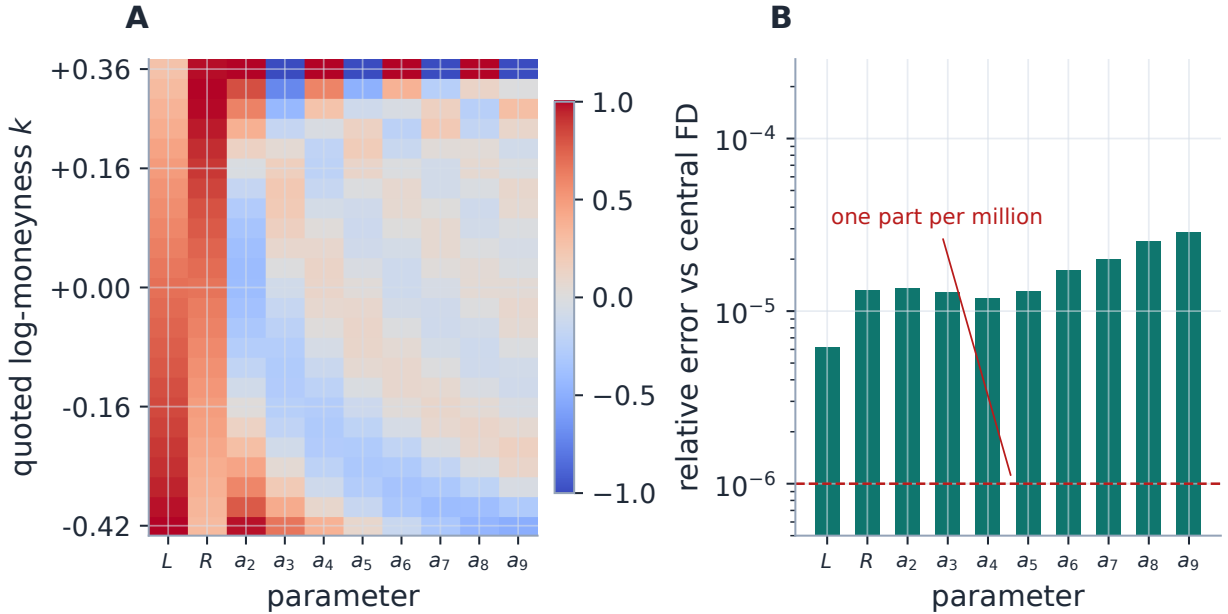


Figure 7: The analytic Jacobian audited against an independent central finite difference on the running fit (10 parameters). Panel A: each column is one parameter’s price sensitivity across the strike ladder, normalized per column to $[-1, 1]$ (the colour scale) — L and R own the wings, and each Legendre mode paints its own oscillation. Panel B: column-wise relative disagreement between the one-pass analytic Jacobian and the central difference: at most 2.88×10^{-5} , i.e. agreement to about five significant digits with a derivative computed by a completely different route.

10.3 Scope and speed

The analytic route covers the var-swap-free, prior-free configuration (table 3); other fits fall back to scipy’s two-point finite difference, correct but not accelerated. Measured on this repository’s development machine, the analytic Jacobian reaches the *same* optimum as the finite difference (costs agreeing to $\sim 10^{-11}$ relative, parameters to $\lesssim 10^{-5}$) at $1.3\text{--}1.7\times$ the speed across $N = 6 \dots 12$ — but wall-clock ratios are machine-, cache- and configuration-dependent, so the measurement protocol, the timing table and the chart live with the other performance engineering in appendix B. The structural point is machine-independent: one quadrature pass now serves all P columns, where the

finite difference rebuilt the pipeline $P + 1$ times.

11 Many expiries: order the ledgers

One dimension of the problem has been silent so far: time. Proposition 1 silences butterflies *within* a slice, but a surface is many slices, and fitting them independently leaves a second static arbitrage on the table: nothing yet prevents a longer-dated call from pricing *below* a shorter-dated one at the same log-moneyness — a calendar spread the market would pay you to own. (Two standing assumptions frame everything here: carry is deterministic, so normalized units under each expiry's forward measure are legitimate simultaneously; and every slice is normalized to mean one.) For expiries $\tau_i < \tau_j$, absence of calendar arbitrage is the requirement $C_i(k) \leq C_j(k)$ at every k — in the language of probability, the laws of Y must increase in *convex order* [5]: the longer-dated law is more spread out, worth at least as much against every convex payoff, without its mean moving.

Checking a continuum of call inequalities sounds like work. The pleasant surprise is that the model already owns the right test object, because the call curve and the upper share are *Legendre transforms of each other*. For any rank u and any log-strike k ,

$$G(u) - e^k(1 - u) = \int_u^1 (e^{Q(v)} - e^k) dv \leq C(k), \quad (44)$$

because extending the integral to where the integrand is negative can only lose money — with equality exactly at $u = u_k$, the strike's own rank. (That equality *is* the pricing identity (25), rediscovered as a maximization.) Read once in each direction:

$$C(k) = \max_{u \in [0,1]} [G(u) - e^k(1 - u)], \quad G(u) = \min_k [C(k) + e^k(1 - u)]. \quad (45)$$

Each curve is an extremum over the other, and pointwise domination passes through maxima and minima in both directions. Hence, for mean-one slices,

$$C_i(k) \leq C_j(k) \text{ for all } k \iff G_i(u) \leq G_j(u) \text{ for all } u \in [0, 1], \quad (46)$$

with equality at $u = 0$, where both shares equal the common mean, one. A continuum of option inequalities has become the pointwise ordering of two curves that are already on the grid — the same ledger that priced the calls and carried the Jacobian, now doing its third job.

Enforcement follows the geometry. The surface is built nearest-to-farthest: each new slice is fitted subject to $G_j(u_m) \geq G_i(u_m)$ on a dense rank grid (per-node slack tolerance zero in production), as the soft calendar block of table 3 with penalty weight `calendarWeight`; the constraint is evaluated at z -values, so it is exact whatever grid the optimizer runs on (section 7.2). Rank space is also the better-conditioned place to test: no implied-volatility inversion at any strike, and full meaning retained in the far wings, exactly where inversion goes blind.

Figure 8 shows the whole story on fitted slices, and it is worth reading deliberately, panel by panel. Panel A draws the ledgers themselves for a legitimate pair: the running example as the far expiry, and a quarter-year slice fitted to the same smile shape with proportionally smaller variance as the near one. Both curves start at $G(0) = 1$ — the whole mean, since every outcome lies above the 0th percentile — and both fall to zero at $u = 1$. In between, the far curve rides everywhere *above* the near one, and the shaded sliver between them is the calendar's entire economic content:

at every rank, the top $(1 - u)$ -tranche of the longer-dated law carries more of the mean, which by (45) is precisely the statement that every call is worth more at the longer expiry. Note how thin the sliver is — a legitimate term structure clears the bar by little, which is why calendar violations are so easy to manufacture and so hard to spot by eye.

Panel B plots that sliver directly: the calendar gap $G_{\text{far}}(u) - G_{\text{near}}(u)$, which the order (46) requires to be non-negative for every rank. The gap must vanish at both ends (at $u = 0$ both ledgers equal the common mean; at $u = 1$ both are empty), so all the information lives in the middle, and three fits are drawn there. Green, the legitimate pair: an arch that stays non-negative everywhere (minimum 0.00087). Red, a deliberately crossed pair — the far slice refitted to quotes whose body variance sits 10% *below* the near slice while its wings sit above: the gap dips to -0.00218 around the $u \approx 0.39$ rank (shaded), and that dip is a concrete trade, a calendar spread at the corresponding strikes that the crossed surface prices *negative*. Each smile on its own looks perfectly respectable; only the ledger comparison exposes the crossing. Dashed blue, the same crossed quotes refitted *with* the production soft floor active: the fit concedes what the quotes demand, the gap is pressed against zero from below, and the residual violation is -3.45×10^{-6} — four orders of magnitude smaller, but not exactly zero, because the floor is a finite-weight hinge, not a hard constraint. That residual is the honest signature of *soft* enforcement, and it is the subject of the closing paragraph.

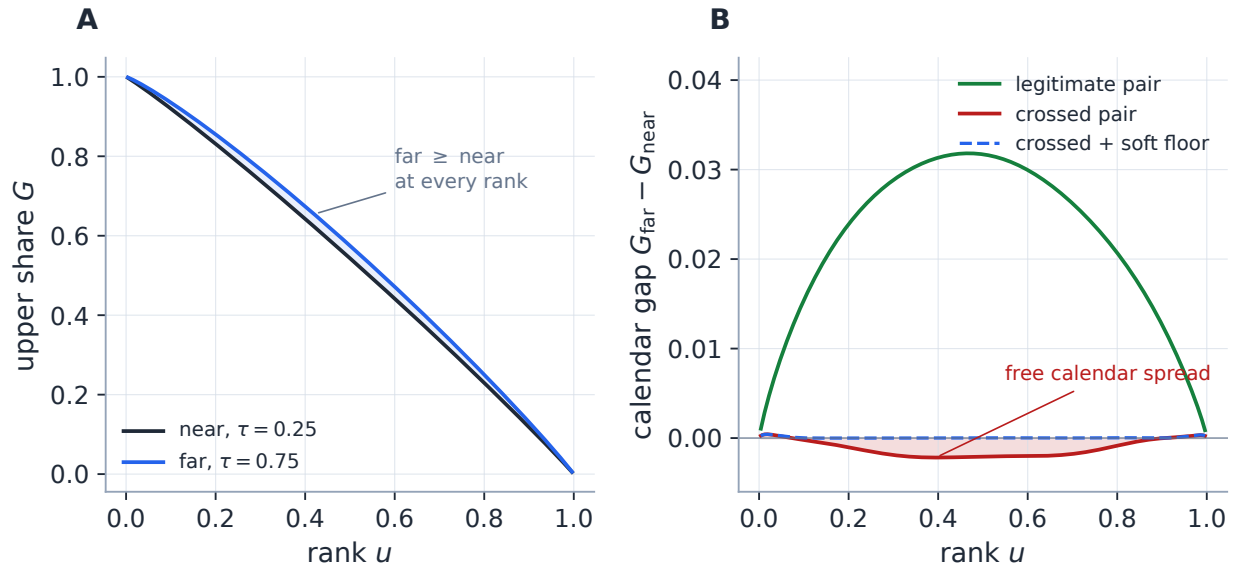


Figure 8: Convex order read off the ledger. Panel A: the upper shares of a legitimate near/far pair; the shaded sliver between the curves — far above near at every rank — is exactly the value ordering of every calendar spread. Panel B: the calendar gap for three fits: the legitimate pair (green, minimum 0.00087), a crafted crossed pair (red; the shaded dip to -0.00218 is a free calendar spread hiding between two respectable-looking smiles), and the crossed quotes refitted under the production soft floor (dashed blue, residual violation -3.45×10^{-6} : pressed to zero, softly).

Honesty about the guarantee: this is *control*, not a theorem. The hinge carries a finite weight, so a hard-pressed fit can retain a small violation rather than destroy the quote fit (the dashed curve of fig. 8 retains exactly -3.45×10^{-6}); the whole coupling is a user toggle (`enforceCalendar`), and off means off; residual violations are measured and reported by the surface diagnostics rather than assumed away. Note 10 develops the model-agnostic calendar machinery (and the subtlety that the

variance floor must be confined to the traded strike range); here we record only that for LQD the natural constraint object is the ledger the model already owns.

12 Case files

A lecture should end with the machine run end-to-end, and with the operator saying plainly what the run does and does not prove. Both cases are fitted by the production calibrator — not a notebook re-implementation — against two deterministic targets; of the note’s generators in Docs/notes/figures, `gen_lqd_fresh.py` computes the quoted diagnostics and `gen_lqd_geometry.py` draws the figures from the identical fits. What they prove is *approximation capacity*: that the basis can reproduce a known clean curve through option prices and back. What they do not prove is market fit; that evidence, against noisy NBBO quotes, lives in the backtest reports (section 1). Table 4 puts the two side by side.

Table 4: The two case files at a glance (fresh production run).

Diagnostic	SPX-like	Event
Model order	9	16
Quotes	24	37
Max error (vol bp)	1.341	2.444
RMS error (vol bp)	0.408	0.999
A_L	0.157473	0.015531
A_R	0.037786	0.014044

12.1 Case file: the SPX-like strip

The running example in full. The target is a skewed SSVI-shaped curve at $\tau = 0.75$ (appendix C) with an ATM volatility near 21.7961% and a pronounced index put skew, sampled at 24 irregular strikes and perturbed by a deterministic sub-two-bp ripple — so the fit is a genuine fit, not a sterile recovery of a noiseless curve. A 9-mode LQD slice converges in 8 residual evaluations with

$$\text{max error} = 1.341 \text{ vol bp}, \quad \text{RMS} = 0.408 \text{ vol bp}, \quad \mathbb{E}[e^X] - 1 = -1.11 \times 10^{-16},$$

exact ATM handles $\sigma_0 = 21.7961\%$, $s_0 = -0.177846$, $\kappa_0 = 0.002673$, endpoint scales $A_L = 0.157473$, $A_R = 0.037786$, Lee slopes $\beta_L = 0.073087$, $\beta_R = 0.019259$, and a native var-swap strike (38) of 22.9685% — above the ATM volatility, as the skew demands. Figure 9 shows the fit, its error ledger, and what the skew looks like as probability; fig. 10 follows the fitted endpoint scales through the maps of section 4.

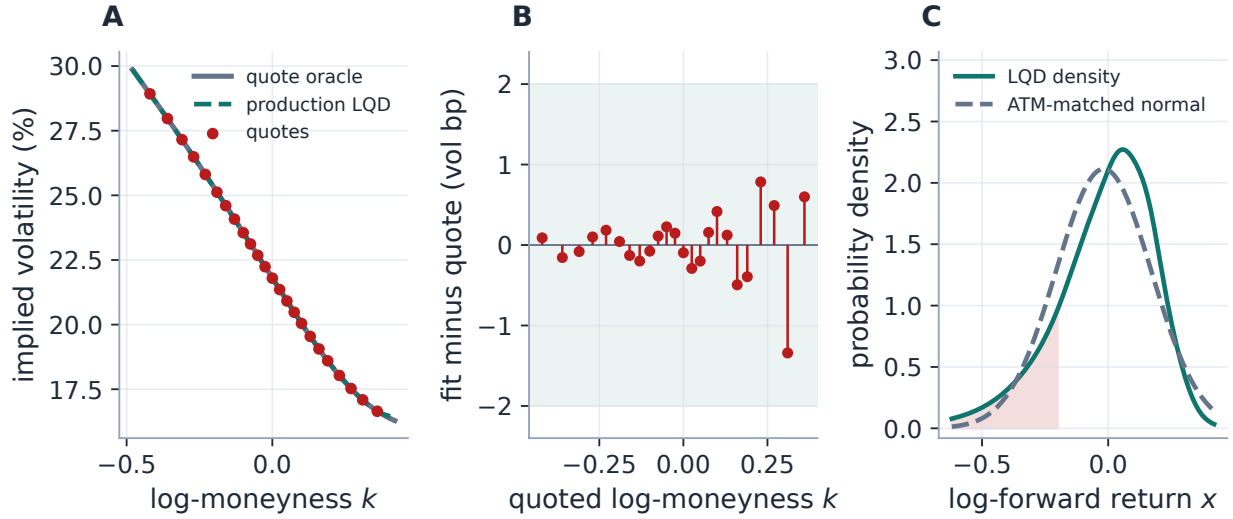


Figure 9: The SPX-like case. Panel A: quote oracle, quotes, and the production fit. Panel B: the error ledger at the quotes — every residual inside ± 2 vol bp (shaded), maximum 1.341 vol bp against quotes carrying a deliberate ~ 2 bp ripple. Panel C: the fitted density against an ATM-matched normal: the entire content of the skew is the transfer of mass into the left tail (shaded), bought by a thinner right shoulder. Beyond the quoted window the curve is the model’s own Lee-consistent extrapolation, not an oracle continuation.

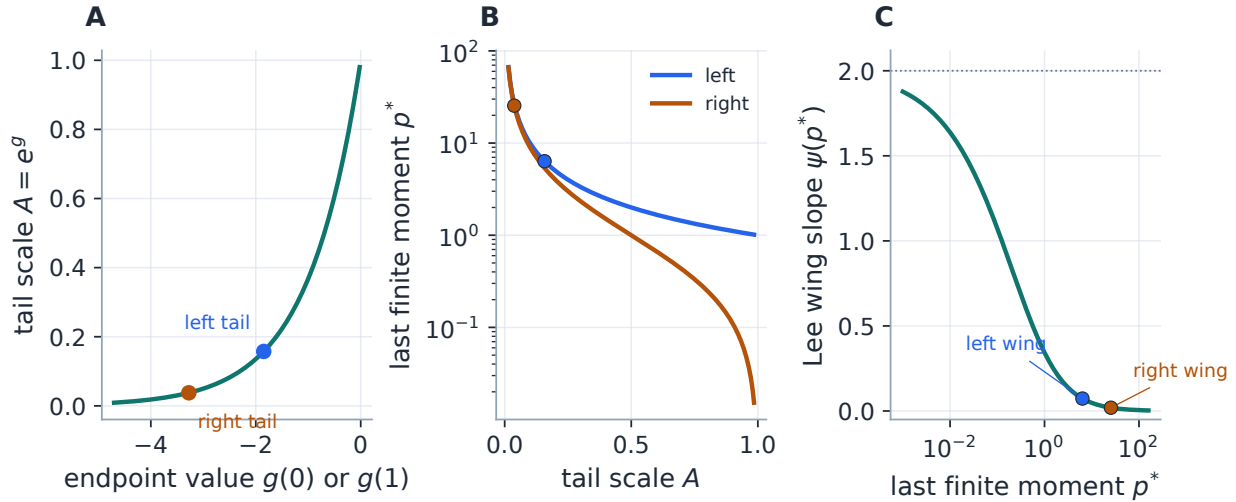


Figure 10: The fitted tails walked through the three links of section 4: endpoint value of g to tail scale ($A_L = 0.157473$, $A_R = 0.037786$), tail scale to last finite moment (13), and moment to Lee wing slope (16) ($\beta_L = 0.073087$, $\beta_R = 0.019259$). Nothing imposed these wings; they are what the quoted body implied through the basis.

12.2 Case file: the density wears two hats

Ahead of a scheduled binary event — earnings, a court ruling, a referendum — the terminal density can be *bimodal*: the market prices two distinct landing zones. A low-order direct-volatility

model fights this shape; a quadratic-in- k smile cannot make a central trough with supported wings. The target here is an asymmetric mixture of two log-return normals (weights 56.0%/44%, centres -0.074804 and $+0.085196$ after martingale normalization, distinct widths; appendix C) at $\tau = 24/365$, quoted at 37 strikes. A 16-mode fit — the top of the schema range, an event slice’s natural habitat — recovers both the smile and the two-humped density (fig. 11) with maximum error 2.444 vol bp, endpoint scales $A_L = 0.015531$, $A_R = 0.014044$, and $\mathbb{E}[e^X] - 1 = -2.22 \times 10^{-16}$. Positivity of the density is automatic even where the high Legendre modes carve the trough; what the basis *cannot* do is place an exact atom — the hats are smooth by construction (section 14). In a sparse live market one would lean on the ridge or fix the handles through the chart of section 8.2 to keep the extra modes from fitting noise.

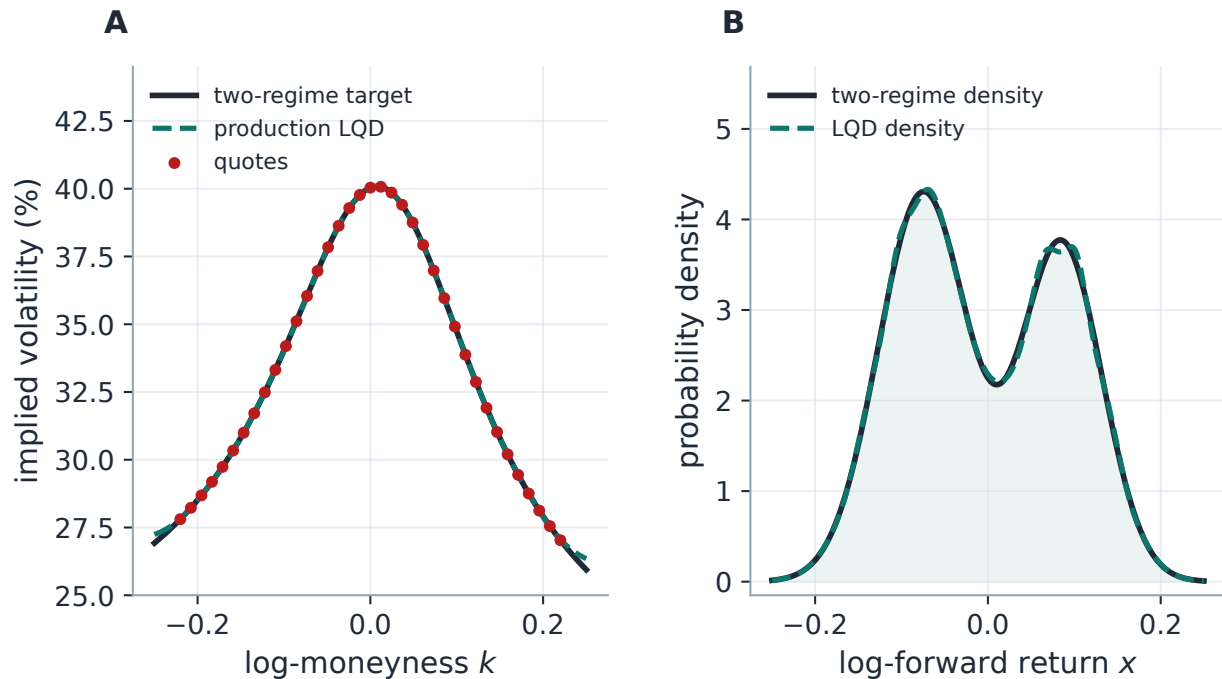


Figure 11: The event case. Panel A: the two-regime target smile and the 16-mode production fit — note the W-shaped body a low-order model cannot make. Panel B: the true mixture density and the LQD density: both hats, the asymmetry, and the trough survive the round trip through option prices, with positivity never at risk.

13 What is genuinely original here

Quantile-based and density-based smile constructions are not new, and the Lee analysis is classical. Three things in this implementation are worth singling out:

1. **The endpoint basis** — $-\log u - \log(1 - u) + (1 - u)L + uR$. It makes the tails exponential *with controllable scale* while keeping the body in a clean orthogonal-polynomial space; the Lee slopes (16) then fall out of the parametrization instead of being patched on.
2. **The exact ATM handles and the orthogonal chart** (section 8). Level, skew and curvature are closed-form functionals of the quantile, so the model carries genuine trader coordinates

and a chart in which shape modes do not disturb them — the same three handles the graph extrapolator of Note 14 propagates.

3. **The implicit-root cancellation** (theorem 1). The strike root’s dependence on θ vanishing from the price gradient turns a $(P+1)$ -rebuild finite-difference Jacobian into a one-pass analytic one, audited to 2.88×10^{-5} column-wise against central differences.

14 Limitations

A lecture that sells a model without drawing its boundary is an advertisement. Here is where this one’s guarantees stop. The headline first: LQD removes butterfly arbitrage by construction but *not* calendar arbitrage across independently fitted expiries — the coupling of section 11 is soft, finite-weight and optional, so a coupled surface is *calendar-controlled*, not *calendar-guaranteed*. Beyond that:

- **Near the integrability wall.** As $A_R \rightarrow 1$ the forward integral degenerates: the tail corrections (26) dominate, conditioning of the fit and of the retarget Newton solve deteriorates, and the barrier is doing real work. Admissible is not the same as comfortable.
- **Atoms of any kind.** The model is a continuous law on $(0, \infty)$: not only a jump-to-default atom at zero but *any* exact mass point (a pinned settlement price, a tender offer) can only be approximated by a sharp continuous feature, never represented.
- **Positive is not plausible.** Structural positivity guarantees a *density*, not a sensible one: an unregularized high-order fit can produce economically implausible — multi-modal, spiky — yet perfectly arbitrage-free shapes. Positivity removes one failure class; the ridge, band weighting and judgment handle the rest.
- **Coefficient non-uniqueness in practice.** Distinct coefficient vectors can price indistinguishably (near-degenerate directions grow with N and with sparse data), so fitted coefficients are not comparable across fits — compare handles, curves and densities, not raw θ .
- **Unstudied sensitivities.** There is no dedicated study of handle *drift* under rolling recalibration, nor of digital/one-touch sensitivities to the interpolation scheme; both are open measurement items, not established properties.

15 Traceability

One caveat on the “match note” anchors: the fixture coefficients those tests lock were recorded when the original benchmark was frozen, so they are *legacy locks* — they pin production behaviour against regression, but their literal values can lag the freshly regenerated numbers in this note (which always come from fresh generator runs). Where the two disagree, the fresh run is the current truth and the test is the regression tripwire.

Table 5: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor <i>Test anchors</i>
Density positive; call curve monotone and convex; parity holds	proposition 1	models/lqd/quadrature.py test_lqd_pricing.py::test_density_positive ::test_call_curve_monotone_and_convex_in_strike ::test_put_call_parity
Martingale shift correct; $\mathbb{E}[e^X] = 1$	(23)	models/lqd/quadrature.py test_lqd_pricing.py::test_svi_fit_martingale_shift_matches_note ::test_svi_fit_martingale_integral_is_one
Endpoint scales and Lee slopes; stable for extreme scales	(7)–(16)	models/lqd/basis.py test_lqd_basis.py::test_svi_fit_endpoint_scales_match_note ::test_svi_fit_lee_slopes_match_note ::test_lee_slopes_stable_for_tiny_nonzero_endpoint_scales
Legendre basis: recursion and orthogonality	(18)	models/lqd/basis.py test_lqd_basis.py::test_legendre_recursion_matches_explicit_polynomials ::test_legendre_orthogonality_on_unit_interval
Exact ATM handles	(33)	models/lqd/atm.py test_lqd_atm.py::test_atm_level_matches_implied_vol ::test_atm_skew_matches_finite_difference ::test_atm_curvature_matches_finite_difference
Orthogonal chart: shape moves preserve handles; retarget exact	section 8.2	models/lqd/ortho.py test_lqd_ortho.py::test_shape_move_leaves_handles_nearly_unchanged ::test_retarget_hits_exact_handles
Analytic Jacobian matches FD (mid, band, calendar); fits agree	theorem 1	models/lqd/jacobian.py test_lqd_jacobian.py::test_jacobian_matches_fd_mid ::test_jacobian_matches_fd_band ::test_jacobian_matches_fd_calendar ::test_analytic_and_fd_fits_agree
Barrier finite for wild trial parameters	(37)	models/lqd/calibrate.py test_lqd_jacobian.py::test_barrier_residual_finite_for_wild_tail_scale
Benchmark fit and speed rail (legacy fixture)	section 12	models/lqd/calibrate.py test_lqd_calibrate.py::test_calibrate_to_svi_benchmark ::test_calibration_is_fast_enough

Claim	Object	Code anchor <i>Test anchors</i>
Double-hat event smile: bimodal density recovered	section 12.2	<code>models/lqd/calibrate.py</code> <code>test_lqd_pricing.py::test_double_hat_density_is_bimodal</code>
Calendar floor exact at constraint z -values	(46)	<code>calib/calendar.py</code> <code>test_lqd_jacobian.py::test_jacobian_matches_fd_calendar</code>

A LQD-native controls and numerical constants

The knobs *specific to* the LQD slice, with defaults and roles. Surfaced knobs are user-facing `FitSettings`; hidden knobs are internal constants tuned to the algorithm and not exposed. The *shared* calibration controls that also shape an LQD fit — `fitMode`, `haircut`, `midAnchorWeight`, `enforceCalendar/calendarWeight`, the var-swap controls, and the prior-persistence and observation-filter settings of table 3 — are indexed in Note 00’s control table and derived in their own notes (07, 08, 10, 13, 15); they are deliberately not duplicated here.

Table 6: LQD-native controls.

Knob	Default	Role
<i>Surfaced (FitSettings)</i>		
<code>nOrder</code> N	6	Legendre expansion order; 7 parameters at $N = 6$. Schema range $4 \leq N \leq 16$: 6 for ordinary slices, 8–14 for scheduled-event slices.
<code>regLambda</code> λ	10^{-6}	High-order ridge coefficient in (36); suppresses oscillation in sparse data.
<code>regPower</code> r	1.0	Ridge exponent in $n^{2r} a_n^2$; larger r penalizes high modes harder.
<code>barrierCenter</code> c	0.90	Centre of the A_R softplus barrier (37); engages before the $A_R = 1$ wall.
<code>barrierScale</code> γ	50.0	Steepness of that barrier.
<code>weightScheme</code>	<code>equal</code>	Per-quote weighting of the data residuals (<code>equal</code> or <code>tv_density</code>); see Note 07.
<i>Hidden (internal constants)</i>		
<code>Z_MAX</code> Z	40.0	Half-width of the symmetric log-odds grid; truncation error $\ll$ vega noise for equity tails.
<code>N_POINTS</code>	8001	Reporting/diagnostic grid nodes (odd, so $z = 0$ is a node).
<code>OPT_N_POINTS</code>	2001	Coarser grid used during optimization iterations; the accepted slice is rebuilt at <code>N_POINTS</code> .
<code>EPS_AR</code> ε_R	10^{-6}	Buffer in the hard bound $A_R < 1 - \varepsilon_R$.

Knob	Default	Role
<code>_VEGA_FLOOR η</code>	10^{-4}	Floor added to Black vega in the residual normalization (28).
<code>xtol/ftol/gtol</code>	10^{-10}	<code>trf</code> tolerances; ~ 6 orders below the fit budget, so the optimum is display-identical while <code>trf</code> stops grinding invisible reductions.
<code>max_nfev</code>	4000	Evaluation cap (rarely approached).

B Performance notes

1. **Cached static grid.** The grid $(z, u, u(1 - u))$ and the Legendre matrix depend only on (Z, n_{pts}, N) , so they are LRU-cached and returned read-only; only g is re-formed per call. This removed $\sim 40\%$ of `build_slice` and made each build $\sim 1.7\times$ faster.
2. **Two-grid optimization.** Iterations run on `OPT_N_POINTS` = 2001 (Simpson error already far below the fit budget); the accepted slice is rebuilt once at 8001 nodes for the reported result and diagnostics. Converged parameters agree with a full-grid fit to $\sim 10^{-6}$.
3. **Hermite interpolation/inversion.** Storing the exact nodal derivatives dQ/dz and dG/dz gives cubic-Hermite off-grid evaluation (clean Greeks) and machine-precision strike inversion by Hermite-Newton — and supplies precisely the nodal sensitivities the analytic Jacobian reuses.
4. **Analytic Jacobian** (section 10). One quadrature pass instead of $P + 1$ finite-difference rebuilds, reaching the same fitted solution within tolerance (relative cost agreement $\sim 10^{-11}$), measured $1.3\text{--}1.7\times$ on the residual-path benchmark of table 7 and structurally more on heavier configurations. Gated to the var-swap/prior-free configuration.
5. **Looser tolerances.** Dropping `trf` tolerances from 10^{-15} to 10^{-10} stops the optimizer chasing cost reductions invisible in the priced surface, saving whole $(P+1)$ -evaluation iterations at no display-precision cost.

The Jacobian timing measurement. Wall-clock speed-ups are machine-dependent, so the numbers promised in section 10 live here with their protocol stated in full. The generator times the least-squares *residual path* (repeated `calibrate_slice` calls on the legacy SVI-JW synthetic benchmark), fastest of three repeats, single-threaded, warm static-grid cache, core residual blocks only; it does not time the public calibrate-and-report workflow, and different hardware, a cold cache, or an active var-swap/prior block will move the ratio.

Table 7: Analytic vs. two-point finite-difference Jacobian, fresh run of the production calibrator under the protocol above. $P = N + 1$ is the parameter count; the cost column confirms the same optimum within tolerance.

N	P	analytic (ms)	FD (ms)	speed-up	$ \Delta\text{cost} $
6	7	15.5	24.7	$1.59\times$	$5.6\text{e-}20$
8	9	18.4	24.9	$1.35\times$	$1.4\text{e-}19$
10	11	25.6	68.8	$2.68\times$	$8.9\text{e-}20$
12	13	30.7	47.6	$1.55\times$	$2.0\text{e-}20$

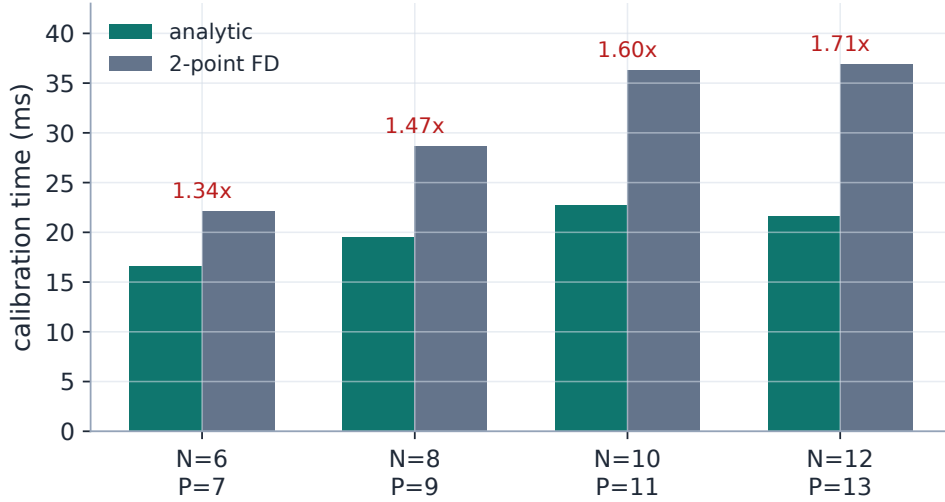


Figure 12: The same measurement drawn: per-fit wall time with the analytic one-pass Jacobian vs the $(P+1)$ -rebuild finite difference, with the ratio above each pair. The win grows with the parameter count, as it must — the analytic pass costs one quadrature regardless of P .

Heuristic

Why only $\sim 1.5\times$ and not $\sim P\times$? Two reasons. First, the cached static grid makes a *repeat build_slice* cheap, so $P + 1$ finite-difference rebuilds cost much less than $P + 1$ cold builds. Second, `trf` converges here in a handful of iterations, so linear algebra and the value evaluation dilute the Jacobian saving. The structural payoff — a Jacobian that no longer rebuilds the quadrature — is what makes higher-order and surface-wide fits comfortable.

C The synthetic oracles

The case files fit production against two deterministic quote oracles; neither is used anywhere inside the model.

The SSVI-shaped strip. The SPX-like target is the total-variance curve

$$w(k) = \frac{\vartheta}{2} \left(1 + \rho \phi k + \sqrt{(\phi k + \rho)^2 + 1 - \rho^2} \right), \quad (\vartheta, \rho, \phi) = (0.0356, -0.68, 2.40), \quad (47)$$

Gatheral–Jacquier’s SSVI slice shape [4], sampled at the 24 strikes listed in the generator and perturbed by the deterministic ripple $10^{-4}(1.20 \sin(9k + 0.4) + 0.55 \cos(21k))$ in volatility. (The regression fixture behind the legacy timing table, table 7, is the older five-parameter raw-SVI benchmark; the JW re-coordinatization used to specify it is Note 02’s subject.)

The two-regime mixture. The event target is an equal-mean mixture of two log-return normals with weights (56.0%, 44%), raw centres $(-0.075, +0.085)$ shifted by a common constant so the mixture is a martingale, and widths $(0.052, 0.047)$. Its normalized calls are the exact two-term Gaussian formula

$$C(k) = \sum_i w_i \left(e^{m_i + \sigma_i^2/2} \Phi\left(\frac{m_i + \sigma_i^2 - k}{\sigma_i}\right) - e^k \Phi\left(\frac{m_i - k}{\sigma_i}\right) \right), \quad (48)$$

inverted to implied volatilities for quoting. This oracle has genuinely bimodal density, which is the point.

D Reference implementation

The whole slice is one quadrature pass in the log-odds coordinate: form g , integrate $dQ/dz = e^g$ to the quantile, solve one scalar normalization for the martingale shift μ , reverse-integrate the upper share G , and price every strike by (25). Executed against production before committing: the listing reproduces `build_slice`’s quantile and share to 10^{-12} and μ to 10^{-14} on an $N = 6$ slice, and returns $\mathbb{E}[e^X] = 1.000000$ (production inverts $Q(z_k) = k$ by Hermite–Newton and interpolates G with cubic Hermite for smooth Greeks; the linear `np.interp` here prices calls to $\sim 10^{-6}$ on the dense grid).

Listing 2: The LQD quadrature pipeline (distilled from `models/lqd/{basis,quadrature}.py`).

```

1  def legendre(n_max, x):                # P_0..P_{n_max}, 3-term recursion
2      P = np.empty((n_max+1, x.size)); P[0] = 1.0; P[1] = x
3      for n in range(1, n_max):
4          P[n+1] = ((2*n+1)*x*P[n] - n*P[n-1]) / (n+1)
5      return P
6
7  def g_of_u(u, L, R, a):                # smooth part g(u) of log q(u)
8      return (1-u)*L + u*R + a @ legendre(len(a)+1, 1 - 2*u)[2:]
9
10 def build_slice(z, L, R, a):
11     # One quadrature pass, eqs (Q_of_z)-(upper_share).
12     u = expit(z); dz = z[1] - z[0]
13     # Tail scales e^{g(0)}, e^{g(1)}:
14     A_L, A_R = np.exp(g_of_u(np.array([0.0, 1.0]), L, R, a))
15     dQ = np.exp(g_of_u(u, L, R, a))      # dQ/dz = e^{g}
16     Q = cumulative_simpson(dQ, dx=dz, initial=0.0)
17     Q -= Q[len(z)//2]                     # anchor Q(0) = 0
18     mass = np.exp(Q)*u*(1-u)
19     tot = (np.trapezoid(mass, z)           # + analytic tails
20            + np.exp(Q[-1]-z[-1])/(1-A_R)
21            + np.exp(Q[0]-z[-1])/(1+A_L))
22     mu = -np.log(tot); Q = Q + mu        # now E[e^X] = 1
23     m = np.exp(Q)*u*(1-u)

```

```

24     # Upper share  $G(z) = \int_z^{-\infty} e^{-\{Q\} u(1-u)} du$ :
25     G = (cumulative_simpson(m[:-1], dx=dz, initial=0.0)[:-1]
26         + np.exp(Q[-1]-z[-1])/(1-A_R))
27     return u, Q, G
28
29 def call_price(k, z, u, Q, G):
30     #  $C(k) = G(z_k) - e^{-k} (1 - u_k)$ 
31     zk = np.interp(k, Q, z) # invert  $Q(z_k) = k$ 
32     return np.interp(zk, z, G) - np.exp(k)*(1 - expit(zk))

```

References

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